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CHEM FILM ALODINE

THE COMPLETE TRAINING MANUAL

CONVERSION COATING · CHEM FILM ON ALUMINUM · CONDUCTIVE · MIL-DTL-5541

*A full training course for the brand-new operator — from "what even is a chem film?" to a signed certificate. The thin **chromate conversion coating** that finishes bare **aluminum** — **chem film, Alodine, or Iridite** — applied by brush, immersion, or spray. Its superpower: it protects aluminum and primes it for paint and stays **electrically conductive**, so the part can be both protected and grounded. Covers **Type I** (gold, hexavalent) and **Type II** (clear, chrome-free). Assumes you know nothing. Teaches you everything.*

01 RECEIVE ▸ 02 CLEAN ▸ 03 RINSE ▸ 04 DEOX

05 RINSE ▸ 06 CHEM FILM ▸ 07 RINSE ▸ 08 DRY·INSP

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · BEGINNER-FRIENDLY STYLE

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW A CHEM FILM FROM A PAPERWEIGHT, YOU'RE IN EXACTLY THE RIGHT PLACE — AND HERE THAT'S ABOUT YOUR HEALTH *AND* WHETHER THE PART CAN CARRY A GROUND

Welcome. You're about to learn how to run — or at least understand — a **chem film line**: the quick chemical process that goes on **bare aluminum parts** to turn raw, freshly cleaned metal into a corrosion-resistant, paint-ready, *and still electrically conductive* finish. You'll hear it called **chem film, chemical film, chromate conversion coating, Alodine,** or **Iridite** — different trade names for the same family. Whether you started this week or have masked aircraft brackets for a year, you're in the right place.

Here is the most important thing we can tell you up front, and we'll say it more than once: **the classic "gold" chem film (Type I) runs on hexavalent chromium — Cr(VI) — a confirmed human carcinogen**, the most hazardous chemistry in the conversion-coating family. The modern **Type II** chemistry replaces it with **trivalent or chrome-free** chemistry that is far lower-hazard — but still an acidic process. The second thing: **this manual assumes you know nothing about chemistry, aluminum, or how a conversion coating works.** That's a promise, not an insult — every term gets defined the first time we use it, in the spirit of the *plain-language* guides: friendly, plain-spoken, and patient.

> ★ **REMEMBER**
> You do not need to memorize this manual. You need to *understand* it. On a chem film line, the understanding is what keeps you healthy *and* keeps the part able to do its job.

> ⚠ **HEADS-UP**
> One big idea before we start: **a chem film line has no electricity running through the parts.** There's no rectifier, no anode, no cathode — the coating forms because the **aluminum surface and the chem film solution react with each other.** That removes the shock-at-the-part hazard, and it's a key difference from *anodizing*, which converts aluminum using electric current. But "no current" does **not** mean "no hazard": you're still working with **acidic deoxidizers, an acidic chromate bath (Cr VI in Type I), alkaline cleaners, and slick wet floors.** Respect all of it.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, in the order a part travels — receive, clean, rinse, deoxidize, rinse, chem film, rinse, dry/inspect. Reading it in order matters: the order of a finishing line is never random, and neither is this book's.

> 💡 **TIP**
> Read Part One all the way through *before* any station chapter — the stations assume you already understand the big picture (the **two jobs** of chem film, the **conductivity** difference from anodize, and the **Cr VI hazard** of Type I) that Part One gives you.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

> 💡 **TIP**
> A shortcut or trick of the trade — what a 20-year veteran would lean over and tell you.

> ★ **REMEMBER**
> A core idea worth locking in — the load-bearing ideas on the page.

> ⚠ **HEADS-UP**
> A safety warning. Read every one — they protect your lungs, skin, and eyes. We use them *very* liberally; Cr VI earns it on the Type I line.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry for the curious. Optional — skip it and you can still run the line.

> ☀ **FUN FACT**
> A bit of trivia to make the science stick (and give you something to say at lunch).

TWO THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (paired with a "Top Mistakes" card) and a bordered **Sign-Off** box your *trainer initials* once you've demonstrated the skill safely. Teach-Back is your study guide; Sign-Off is the gate.

FIND YOUR WAY

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★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, your shop’s OSHA Cr(VI) program (Type I), the governing spec, or your supervisor. When this manual and your shop’s official procedure disagree, **follow your shop’s procedure and ask your supervisor.**

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS FOUNDATION AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what a conversion coating is** in plain language — and how it differs from electroplating and anodizing (chemistry alone, *no electric current*).
2. **Describe chem film's TWO defining jobs** — a thin chromate conversion film on bare aluminum that (a) gives **corrosion protection** and an excellent **paint/primer and adhesive base**, *and* (b) uniquely **stays electrically conductive** (low contact resistance), so the part can still be grounded or bonded.
3. **State the conductivity differentiator vs anodizing** — chem film is thin and **conductive**; anodize is thicker, harder, and **electrically insulating.** That's why electronics enclosures, RF/EMI grounding, and aircraft electrical bonds call out chem film.
4. **Name the two Types and the Classes of MIL-DTL-5541** — **Type I** = hexavalent (gold/tan, highest bare corrosion, Cr VI); **Type II** = trivalent / chrome-free (clear/pale, RoHS-driven). **Class 1A** = maximum corrosion protection; **Class 3** = low electrical resistance (thinner, for bonds).
5. **State the central safety fact of the Type I line:** the gold bath contains **hexavalent chromium (Cr VI), a confirmed human carcinogen**, and name the controls that keep you safe (ventilation, mist suppression, respiratory protection, hygiene, $Cr^{6+} \rightarrow Cr^{3+}$ waste treatment).
6. **Explain the regulatory driver** — that **RoHS and REACH restrict hexavalent chromium**, which is why the industry is moving to **Type II** trivalent / chrome-free chem film.
7. **Name the stations of the line in order** — especially the **critical deoxidize/desmut step** — and explain *why* the order can't be rearranged.
8. **Apply the key operator process lessons** — a clean, fully deoxidized surface is everything; pull the chem film at the target color/coating weight; *do not dry the fresh film hot.*
9. **Read color as coverage** — gold/tan Type I confirms coverage; clear Type II is hard to see, so **coverage verification matters more.**
10. **Spot trouble early** — no color, powdery film, **high contact resistance**, poor salt spray — and talk the language with your lead, your chemist, and your supplier.

> ★ **REMEMBER**
> You are not expected to hit all of these on day one. Surface finishing is a craft. But the *safety* objectives (#5) are not optional and not gradual on a Type I line — you must have those before you work the line at all.

• THE EIGHT-STATION PROCESS AT A GLANCE

The whole chem film line in one strip. Don't memorize it yet — just feel the rhythm: **Receive → Clean → Rinse → DEOX → Rinse → CHEM FILM → Rinse → Dry (cool) / Inspect.**

01	02	03	04	05	06	07	08
RECEIVE	**CLEAN**	**RINSE**	**DEOX**	**RINSE**	**CHEM FILM**	**RINSE**	**DRY/INSPECT**
alum in	alkaline	overflow	strip oxide · critical	cool	Cr VI / Cr-free	cool, gentle	color & resistance

> ★ **REMEMBER**
> Notice the pattern: **a rinse follows each chemical stage**, the **deox is its own critical station** (aluminum's natural oxide *must* be stripped or the film won't form), and the **dry-off is deliberately *cool*, not hot.** A hot oven would *destroy* the soft gel film you just grew — the same don't-over-dry rule you'd find on the zinc chromate line.

> ⚠ **HEADS-UP**
> This eight-station layout is the *teaching* version. Your shop's real line may add or combine steps (a separate etch clean, a desmut after an etch, a dragout-recovery rinse, a deionized final rinse, or a Type II process run warm at higher pH). Always run *your* line's actual posted sequence and parameters.

CHAPTER 1

WHAT IS A CONVERSION COATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT "FINISHING" MEANS

THE ONE-SENTENCE VERSION

A conversion coating is a protective film you grow on a metal surface by making the metal itself react with a chemical solution — turning the top layer of the metal *into* the coating. That's the whole idea. Now let's unpack it, because the word "conversion" is doing a lot of work.

THREE COUSINS: PLATING, ANODIZING, AND CONVERSION COATING

It's worth lining all three up side by side, because the differences are the key to understanding chem film — *especially* the difference from anodizing, which is chem film's closest competitor on aluminum.

- **Electroplating** *adds* a layer of a *different* metal on top of your part, using **electricity.**
- **Anodizing** *converts* the part's *own* aluminum surface into a thick, hard oxide — but it uses **electricity** to drive the reaction, and the result is **electrically insulating.**
- **Conversion coating** (this manual) *converts* the part's own surface into a thin protective film using **only chemistry — no electricity at all.** You clean, deoxidize, and dip (or brush/spray) the part, and a new chromate film forms right where the bare aluminum used to be — **and it stays conductive.**

> ★ **REMEMBER**
>
> Lock in the difference: **Plating adds a new metal with electricity. Anodizing converts aluminum with electricity and ends up insulating. Chem film converts aluminum with *chemistry alone* — no rectifier, no anode, no cathode, no current — and stays electrically conductive.** If you ever go looking for the "power supply" feeding the parts on a chem film line, you won't find one. That's not a missing piece — that's the whole point, and it's the headline difference from anodize.

WHERE THIS ONE FITS: THE FINISH *ON* THE ALUMINUM ITSELF

Here's what makes chem film different from the zinc chromate in our other manuals: **the surface it converts is the bare aluminum part itself** — not an electroplated layer on top of steel. A chem film reacts directly with **aluminum** (and aluminum alloys). So a chem film line doesn't sit at the tail of a plating line; it's a **standalone finishing line for aluminum parts** — aircraft brackets, electronic chassis, housings, heat sinks, RF enclosures — that arrive as bare machined or formed aluminum and leave protected, paintable, and still able to carry a ground.

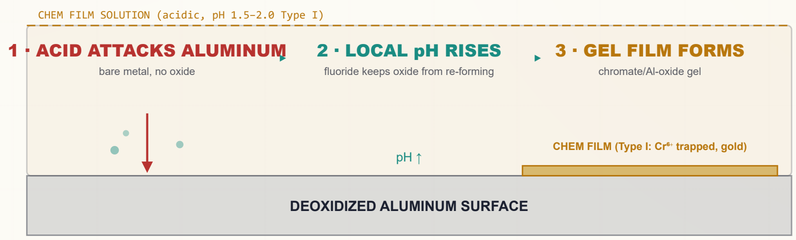
> ☀ **FUN FACT**
>
> Chromate conversion coatings for aluminum trace to the 1940s–1950s, and the trade name **"Alodine"** (from Henkel/Amchem) became so common that operators say "alodine the part" the way people say "xerox a page." **"Iridite"** (MacDermid) is another long-running trade name for the same family. The generic term is **chem film** or **chemical film** — and the governing U.S. spec is **MIL-DTL-5541.**

HOW A CONVERSION COATING FORMS — NO CURRENT REQUIRED

So if there's no electricity, what makes the coating appear? The metal and the solution do it themselves. Here's the plain-language version for chem film on aluminum:

1. First, the aluminum's tenacious **natural oxide must already be stripped** (that's the deox step — Chapter 4 / Station 04). On a *fresh, oxide-free* aluminum surface, the **acidic chem film solution** begins to react with the metal.
2. The acid attacks the aluminum slightly; the **local pH at the surface climbs**, and fluoride activators keep breaking down any re-forming oxide so the reaction can continue.
3. Dissolved **chromium and the reaction products precipitate** as a thin, gel-like **chromate/aluminum-oxide film** right onto the part.
4. In **Type I**, that film **traps a reservoir of soluble hexavalent chromium** — which gives it both its gold color *and* its self-healing (hold that thought for Chapter 2).

HOW THE FILM FORMS (NO ELECTRICAL CELL)



No electricity — the aluminum and the solution do it themselves.

⚙ TECHNICAL STUFF

A classic **Type I** bath (e.g., the Alodine 1200-series chemistry) is essentially **chromic acid + fluoride activators + a ferricyanide accelerator**, pH ~1.5–2.0. The fluoride keeps stripping the ever-re-forming aluminum oxide so the reaction can proceed; the ferricyanide accelerates film growth and contributes to the **gold/tan color.** The film that precipitates is a hydrated **chromium(III) oxide backbone holding a reservoir of soluble chromium(VI)** — a mixed film, just like the zinc version. **Type II** chemistries replace the Cr(VI) with **trivalent chromium (a "TCP")** or other chrome-free chemistry, usually run at a **higher pH (~3.5–4.0) and sometimes mildly warm** — follow the TDS.

⚠ HEADS-UP

"No electricity at the part" removes the shock-at-the-tank hazard — but on a **Type I** line the chem film bath is a **carcinogenic hexavalent-chromium solution**, the deoxidizer is an **acid**, and the line's heaters, pumps, hoists, and dryer are still serious electrical equipment. **Any maintenance still requires Lockout/Tagout (LOTO).** No current through the parts does not mean no hazard in the building.

CHAPTER 2

WHAT IS CHEM FILM?

THE THIN CHROMATE FILM THAT PROTECTS ALUMINUM, PRIMES IT FOR PAINT — AND *KEEPS IT CONDUCTIVE*

CHEM FILM'S BIG JOBS: PROTECT, BOND, CONDUCT (AND, FOR TYPE I, HEAL)

Here's the headline, and it's one of the most important ideas in this manual: **chem film is the conversion coating that takes a bare aluminum part and makes it corrosion-resistant and paint-ready — while keeping it electrically conductive.** That last part is its superpower. It does several jobs at once:

1. **Corrosion protection.** Bare aluminum forms a thin natural oxide, but it still pits and corrodes — especially high-strength alloys and in salt environments. A chem film dramatically improves corrosion resistance.
2. **Paint / primer / adhesive bonding base.** This is *huge* for aerospace. Chem film is the standard pretreatment under primer and paint on aluminum, and under structural adhesives — it gives them a chemically active, well-bonded surface to grip.
3. **Electrical conductivity / low contact resistance. Uniquely among aluminum finishes, chem film protects the metal while still letting electricity pass through the surface.** That's why it's specified for electronics enclosures, RF/EMI shielding and grounding, ground planes, and aircraft electrical bonds. (See **Class 3**, below.)
4. **Self-healing (Type I only).** Like the zinc hex chromate, **Type I** chem film traps mobile Cr(VI) that can re-protect a scratch or a machined/cut edge. **Type II** chemistries largely lack this.

> ★ **REMEMBER**
> Chem film is the only common aluminum finish that does the **protect-and-paint-base** job *and* the **stay-conductive** job at the same time. If a print says the aluminum part must be corrosion-protected **and** electrically grounded/bonded, the answer is almost always **chem film** — not anodize (which insulates).

THE SIGNATURE DIFFERENCE: IT STAYS CONDUCTIVE

Why does conductivity matter so much? Because the obvious alternative for aluminum — **anodizing** — grows a thick, hard **oxide that is an electrical insulator (a dielectric).** Anodize is wonderful for wear and corrosion, but you **cannot run a ground or an RF bond through it.** Chem film's film is so **thin** that current still passes — so a chassis can be both finished *and* electrically continuous. On aircraft, faying surfaces and bonding straps are chem-filmed precisely so the structure stays electrically bonded (for lightning protection, static dissipation, and avionics grounding).

> ★ **REMEMBER**
> **Chem film = protected + conductive. Anodize = protected + insulating.** That one sentence drives most "chem film vs anodize" decisions you'll see on a print.

THE SIGNATURE TRICK (TYPE I): SELF-HEALING

Why is **Type I** chem film so corrosion-tough for such a thin film? Partly barrier, but mostly **self-healing.** The gel film traps a reservoir of soluble **hexavalent chromium.** When the film is later **scratched, machined, or cut** and that bare spot gets wet, the trapped Cr^{6+} **leaches out, migrates to the exposed aluminum, and re-passivates it** — chemically re-forming protective chromate over the damage. The coating, in effect, **patches its own wounds.** (Deep dive in Part Three.)

> ★ **REMEMBER**
> **Self-healing = the soluble hexavalent chromium in the film leaching out to re-protect a scratch.** It's *the* reason Type I gold chem film has been valued for so long — and, because it depends on mobile Cr(VI), it's exactly the property the **Type II** (trivalent/Cr-free) chemistries cannot fully reproduce. That trade-off is the heart of the Type I vs Type II story (Chapter 3 / Part Three).

> ☀ **FUN FACT**
> That self-healing behavior is called "active corrosion protection." A scratched anodized part just relies on its barrier; a scratched **Type I** chem-filmed part can actually re-heal the scratch. It's one of the few coatings that fights back — which is why hex chem film hung on in aerospace long after most commercial finishes went chrome-free.

• TYPES AND CLASSES — READ THE PRINT CAREFULLY

MIL-DTL-5541 sorts chem film two ways at once — by **Type** (the chemistry) and by **Class** (the job). You must know which the print calls out.

TYPE	CHEMISTRY	LOOK	CORROSION	NOTE
Type I	**Hexavalent chromium (Cr VI)**	Gold / tan iridescent	**Highest** bare protection	Classic “gold Alodine 1200”; **carcinogen**; aerospace legacy
Type II	**Trivalent (TCP) or chrome-free**	Clear to faint blue/iridescent	Good, improving	RoHS/REACH-driven; e.g. trivalent-chromium-process products

CLASS	FOR	FILM WEIGHT	NOTE
Class 1A	**Maximum corrosion protection** — painted or unpainted	Heavier	Full coating; paint base; best salt-spray
Class 3	**Low electrical resistance** — electrical/electronic bonds	**Lighter / thinner**	Keeps contact resistance low for grounding

★ **REMEMBER**

Type = the chemistry (I hex / II chrome-free). Class = the job (1A max corrosion / 3 low electrical resistance). A callout like “chem film per MIL-DTL-5541, Type II, Class 3” tells you *both* the chemistry (trivalent/Cr-free) *and* the job (a thin, low-resistance film for an electrical bond). Read both halves. Don’t run a heavy Class 1A film where the print wants Class 3 — you’ll blow the contact-resistance requirement.

⚙ **TECHNICAL STUFF**

Class 3 is thinner *on purpose:* a heavier chromate film raises **electrical contact resistance.** MIL-DTL-5541 sets a maximum contact resistance for Class 3 (commonly cited on the order of **5,000 microohms per square inch as-applied**, with a higher allowance after salt-spray exposure) — **verify the exact value and test conditions against the current revision of the spec**, because the numbers and pressures (e.g., 200 psi) are spelled out there and I won’t quote them as gospel from memory. **Class 1A** has no low-resistance requirement, so it can run heavier for maximum corrosion life.

COLOR — AND WHY IT READS DIFFERENTLY ON TYPE I VS TYPE II

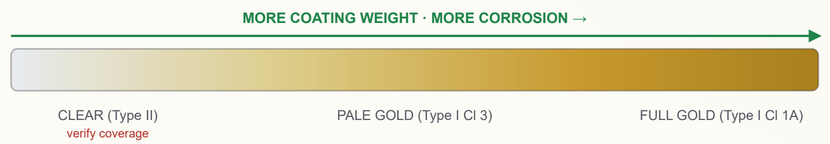
A chem film's color comes from **film thickness/coating weight** and, in Type I, the **trapped hexavalent chromium** (the ferricyanide-accelerated gold).

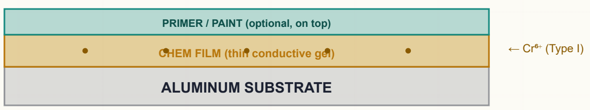
TYPE / CLASS	LOOK	WHAT THE COLOR TELLS YOU
Type I, Class 1A	**Full gold / tan iridescent**	Heavy film, good coverage, high corrosion + self-healing reservoir
Type I, Class 3	**Light gold / pale straw**	Thin film for low resistance — lighter color is *expected*
Type II (triv./Cr-free)	**Clear to faint blue/iridescent**	Very hard to see → **coverage must be verified deliberately**

★ **REMEMBER**

Two ideas. **(1) On Type I, color is a coverage *and* protection signal** — a bright, even gold says the film formed and the Cr(VI) reservoir is present; a pale, patchy gold can mean thin or incomplete coverage. **(2) On Type II, you mostly *can't* read it by eye** — the film is nearly colorless, so you cannot rely on "looks gold = covered." Coverage verification (wet-film/water-break or witness checks per your shop) becomes essential. Always confirm the **exact Type/Class the print calls out.**

READING CHEM FILM BY COLOR · AND WHERE THE FILM SITS





The only aluminum finish you can both paint and ground.

⚙ **TECHNICAL STUFF**

The gold/tan of Type I is largely the inherent color of the **entrapped Cr(VI)** plus the **ferricyanide** reaction product, with some thin-film interference (soap-bubble physics). One practical consequence: because the gold *is* partly the Cr(VI) reservoir, **a strong, even gold is a visual cue the self-healing reservoir is present**, and a washed-out gold can mean it was leached or over-rinsed. Type II has no such reservoir to color it — hence nearly clear.

• WHERE CHEM FILM WORKS — THE SUBSTRATES

Chem film needs a reactive, freshly cleaned and **deoxidized aluminum** surface to convert. It works on:

- **Wrought aluminum alloys** — 6061, 5052, 7075, 2024, 3003, and many others (the everyday aerospace and electronics alloys).
- **Cast aluminum** — works, but high-silicon casting alloys can be harder and may need a fluoride-containing deox/desmut.
- **High-copper alloys (e.g., 2024, 7075)** — chem-film well but leave heavy **copper smut** during cleaning that *must* be desmutted, or you get poor, blotchy film.

> ⚠ **HEADS-UP**
> **A different base metal needs a different conversion coating.** Chem film is for **aluminum.** The chromate that finishes **zinc-plated steel** (clear/yellow/olive/black "passivate," ASTM B633) is a *separate process* with its own chemistry and its own manual. If **steel, zinc-plated, or anodized** parts show up on the chem film line, **flag them** — they don't belong here.

• WHERE CHEM FILM FITS IN THE SPECS

SPEC	WHAT IT COVERS	NOTE
MIL-DTL-5541	The **applied** chromate conversion coating on aluminum — **Type I/II, Class 1A/3**	The master spec; what most prints call out
MIL-DTL-81706	The **chemical materials** used to produce chem film (qualifies the products)	Classifies the chemicals by Type/Form/Method — **verify designations against the current revision**
AMS-C-5541	SAE/aerospace material-spec version of chem film	Commercial-equivalent callout; same Type/Class concept

> ★ **REMEMBER**
> **MIL-DTL-5541 covers the *coating on the part*; MIL-DTL-81706 covers the *chemicals* used to make it.** A product is "qualified to 81706" (the chemistry is approved) and the finished part is "processed to 5541" (the applied coating meets requirements). Prints usually call out **5541, with a Type and a Class.**

• A FIRST LOOK AT THE PROCESS NUMBERS

PARAMETER	TYPICAL RANGE (VERIFY TDS)	WHY
Bath type (Type I)	Acidic **Cr(VI)** (chromic acid) + fluoride + ferricyanide	This trio converts aluminum and colors it gold
Bath type (Type II)	**Trivalent (TCP) / chrome-free**, milder	RoHS/REACH-compliant
pH	`Type I ~1.5–2.0; Type II ~3.5–4.0`	Acid drives the reaction; Type II runs milder
Temperature	`Room temp; some Type II mildly warm`	Unlike anodize, no heated H_2SO_4 tank
Immersion time	`~1–5 min (brush 1–3 min)`	Longer than zinc chromate's seconds; over-time powders the film
Drying	`Low temp — below ~60°C (140°F)`	A hot dry **cracks the soft gel** and (Type I) destroys self-healing

> ★ **REMEMBER**
> Three things define the chem film step: it's usually **cool (or only mildly warm), it's a *few minutes* (not seconds, not an hour), and the part must be dried *gently*.** If your instinct from an anodize or paint-bake line is "hotter and longer is better," unlearn it here — on a chem film, a hot bake *destroys* the coating.

CHAPTER 3 · THE MOST IMPORTANT CHAPTER

WHY HEXAVALENT?

THE CR(VI) HAZARD AND THE ROHS DRIVER, UP FRONT — READ THIS CHAPTER TWICE

We put this chapter near the front on purpose. The classic **Type I** chem film runs on **hexavalent chromium, Cr(VI)** — a **confirmed human carcinogen** — and that is also the single biggest reason the industry is moving to **Type II.** You need both — the *health* hazard and the *regulatory* hazard — before you read another word about how to run the line.

WHY THE CLASSIC BATH IS HEXAVALENT (AND WHY THAT'S THE PROBLEM)

The Type I bath gets its power and its gold color from **hexavalent chromium** (the +6 oxidation state, from chromic acid, CrO_3). That Cr(VI) is what makes the film form, gives it the gold color, *and* gives it self-healing (the trapped Cr^{6+} reservoir). **The very property that makes the coating work is what makes it dangerous:** mobile, soluble, oxidizing hexavalent chromium.

TECHNICAL STUFF

"Hexavalent" (Cr^{6+}, Cr VI) and "trivalent" (Cr^{3+}, Cr III) describe the chromium atom's oxidation state. **Cr(VI) is a strong oxidizer, highly toxic, and carcinogenic. Cr(III)** — the form chromium ends up in *after* it's reduced — is far less hazardous and is even a trace nutrient. **The goal of waste treatment (Part Three) is to convert dangerous Cr^{6+} into safe Cr^{3+} and remove it.** And the goal of the **Type II** chemistry is to build the film out of Cr(III) (or no chromium) from the start.

HOW CR(VI) HURTS PEOPLE

This is not abstract. Cr(VI) exposure on a finishing line causes real, documented harm:

- **Lung and sinonasal cancer** — the most serious; from long-term inhalation of chromic-acid mist.
- **Nasal septum ulceration and perforation** — chromic-acid mist can eat through the cartilage between the nostrils. Preventable with proper controls.
- **"Chrome holes" / chrome ulcers** — deep, slow-healing skin ulcers where chromic acid meets even a tiny break in the skin.
- **Skin and respiratory irritation, dermatitis, allergic sensitization.**
- **Eye damage** — chromic acid is corrosive to the eyes.

HEADS-UP — THE HAZARD IS MOSTLY INVISIBLE

Acidic chromate solution and its **mist/aerosol** are the main route into your lungs, and you can be over-exposed without seeing or smelling much. **Engineering controls (ventilation + mist suppression) are your primary protection — not just PPE.** If ventilation or mist control isn't working on a Type I tank, it's not safe to run. Stop and report it.

THE LAW (PART 1): OSHA'S CR(VI) STANDARD

Cr(VI) has its own dedicated OSHA standard — **29 CFR 1910.1026**. Know the numbers your shop's program is built around:

TERM	VALUE	PLAIN MEANING
PEL (Permissible Exposure Limit)	5 µg/m³	The legal ceiling for Cr(VI) in the air you breathe, averaged over an 8-hr shift
Action Level	2.5 µg/m³	The "pay attention" trigger (8-hr avg) — at/above this, the shop must do air monitoring, training, and medical surveillance

A microgram (µg) is a *millionth* of a gram. A limit of 5 µg per cubic meter of air is *extraordinarily* small — which tells you how potent this hazard is and why mist control is non-negotiable.

⚙ **TECHNICAL STUFF**

Because the PEL is so low, you **cannot** judge your exposure by smell or irritation — those happen at far higher levels. That's why OSHA requires *air monitoring*, *medical surveillance*, regulated areas, a written exposure-control plan, and specific hygiene facilities. Your shop's Cr(VI) program is a legal requirement, not a suggestion.

THE LAW (PART 2): THE ROHS / REACH DRIVER

A second body of law restricts hexavalent chromium **in finished products and on the market:**

- **RoHS** (Restriction of Hazardous Substances) — restricts Cr(VI) in **electrical and electronic equipment.**
- **REACH** — places hexavalent-chromium substances on the EU **authorization** list, so their use requires special, time-limited authorization and is being actively driven down.
- (For automotive, the **ELV** directive restricts Cr(VI) too.)

The combined effect: **for most commercial markets — electronics, consumer goods, much of automotive — hexavalent chem film is increasingly unacceptable.** This is *the* reason the industry developed **Type II trivalent / chrome-free** chem film.

★ **REMEMBER**

Two legal stories, same direction. **OSHA 1910.1026** protects the *worker* from the carcinogen in the tank. **RoHS / REACH** restrict the carcinogen in the *product.* Together they're why **Type II** is taking over — and why aerospace, which still leans on Type I for its self-healing and existing qualifications, is steadily qualifying Type II (TCP) replacements.

⚠ **HEADS-UP**

Knowing Type II *exists* does **not** change how you must treat a Type I tank in front of you. If your shop runs gold (Type I) chem film, it's Cr(VI), a carcinogen. Treat it that way, every shift.

• THE FIVE PILLARS OF CR(VI) CONTROL (TYPE I)

You'll see these at every Type I station:

1. **Mist suppression at the tank** — a **fume suppressant** and/or floating covers. First line of defense, right at the source.
2. **Tank ventilation** — **local exhaust** (lateral or push-pull hoods) pulling mist off the surface, often with a **scrubber** that washes Cr(VI) out of the exhaust air.
3. **Respiratory protection** — the right respirator per your shop's program, used *in addition to* (never instead of) ventilation.
4. **Hygiene and no hand-to-mouth** — **no eating, drinking, smoking, vaping, cosmetics** near the line; designated wash-up; separate work clothing; decon before breaks.
5. **Spill, waste, and skin protection** — full chemical PPE, immediate flushing of any contact, controlled spill response, and proper **Cr(VI) waste treatment** (reduce $Cr^{6+} \rightarrow Cr^{3+}$, then precipitate — Part Three).

> ★ **REMEMBER**
> Engineering controls first (suppress the mist, ventilate the tank), respirator second, hygiene always. **PPE is the *last* layer of defense, not the first.** A respirator does not make a tank with broken ventilation safe.

• HONEST POSITIONING: TYPE I VS TYPE II

	TYPE I — HEXAVALENT (THIS LINE)	TYPE II — TRIVALENT / CHROME-FREE
Chromium state	**Cr(VI) — carcinogen**	**Cr(III) or none — far safer**
Self-healing	**Yes — strong** (mobile Cr^{6+})	Weak / largely absent
Color	Gold/tan (easy to verify)	Clear/pale (harder to verify)
Bare corrosion	**Highest** (classic)	Good; modern TCP close to Type I
Conductivity (Class 3)	Yes	Yes
RoHS / REACH	**Restricted**	**Compliant** — the modern standard
Status	Hangs on in **aerospace/defense** (qualifications, self-healing)	**Default** for new commercial/electronics; growing in aerospace

> ★ **REMEMBER**
> Frame it honestly: **Type I gold gives the best classic corrosion + self-healing and is easy to verify by color — but it's a carcinogen and RoHS/REACH-restricted.** Type II is the compliant, lower-hazard modern choice that's harder to see and (historically) slightly less corrosion-tough, though modern chemistries have nearly closed the gap. Both are legitimate to *know*; the industry direction is Type II.

> ⚠ **HEADS-UP**
> If you take only one thing from this chapter: a **Type I** chem film bath is a **confirmed human carcinogen** whose worst hazard is an *invisible mist*, and it's a **regulated substance** the world is phasing out. Suppress the mist, ventilate the tank, wear your respirator, keep your hands away from your face, and never let Cr(VI) reach a drain. Do those things, every shift, and you can work this line safely.

CHAPTER 4

HOW A CHEM FILM LINE WORKS

A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A chem film line is a **sequence of stations** a part travels through in a strict order. Because chem film converts the bare metal directly, the line starts with raw **aluminum** parts — machined, formed, or fabricated — that must be made chemically clean and **oxide-free** before the film will form. Parts are typically **rack-processed** (hung or fixtured), **barrel-processed** for small hardware, or **brush/spray-applied** for large structures and field repair.

> 💡 **TIP**
> The big mental model for this line: **everything before the chem film exists to deliver a chemically *clean, oxide-free, active* aluminum surface; everything after exists to *protect the soft film you just grew until it cures.*** And the make-or-break pretreatment is the **deox** — aluminum's natural oxide is tenacious, and the film simply won't form on top of it.

• THE WHOLE CHEM FILM SEQUENCE (TEACHING LAYOUT)

#	STATION	THE ONE THING IT DOES	TEMP	NOTES
01	Receive & Inspect	Confirm **bare aluminum**, right alloy/spec, good condition	`Ambient`	Wrong substrate (steel, anodized) → flag it
02	Alkaline Clean / Degrease	Remove oils, fingerprints, shop soil	`Often warm`	Usually **non-etch/mild**; some lines etch then desmut
03	Rinse (post-clean)	Wash off cleaner	`Ambient`	Firewall before the acid
04	**Deoxidize / Desmut**	**Strip the natural oxide + alloying smut — essential**	`Ambient`	**Acidic; the film won't form on heavy oxide**
05	Rinse (post-deox)	Wash off deox acid	`Ambient`	Firewall before the chem film
06	**Chem Film / Alodine**	**Grow the chromate conversion film**	`Amb / mild`	**Main event; Type I = Cr(VI); ~1–5 min; pull at color/Class**
07	Rinse (gentle / cool)	Rinse the soft film without stripping it	`Cool`	Over-hot/over-long rinse leaches protection (Type I)
08	Dry-Off (low temp) & Inspect	Dry **gently** so the gel cures; verify color/coverage/resistance	`≤~60°C`	**Do NOT over-dry — the key operator rule**

Basic rhythm: **Receive → Clean → Rinse → DEOX → Rinse → CHEM FILM → Rinse → Dry (cool) / Inspect.** Note: chem film usually needs **no seal** — it's left bare or painted directly.

> ⚠ **HEADS-UP**
> Notice the bath temperatures: chem film runs **cool or only mildly warm** — *not* the hot sulfuric tank of anodizing. And notice the **dry-off is deliberately *low temperature.*** Both are easy to get wrong if you're used to an anodize or paint-bake line. **Cool/mild process, gentle dry.**

• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

★ **REMEMBER**

Every station either protects the next or poisons it. There is no neutral step. A part that isn't fully **deoxidized** won't take an even film (the oxide blocks it). A part that drags cleaner into the deox, or deox acid into the chem film, shifts the chemistry. A part rinsed too hot loses its Type I self-healing reservoir. A part dried too hot cracks before it cures. What you do — or fail to do — early shows up as a corrosion, conductivity, or adhesion failure later.

💡 **TIP**

Think of the chem film tank as the crown jewel of the line, and the **deox** as the gatekeeper standing in front of it. Everything *before* the chem film delivers a clean, **oxide-free**, active aluminum surface into it. Everything *after* protects and cures the soft film it created. The whole line orbits that one tank — but it only works if the deox did its job.

• WHY A FULLY DEOXIDIZED SURFACE MATTERS

Aluminum is special: the instant it meets air, it grows a thin, tough, invisible **oxide layer** — and that oxide is a near-perfect *barrier* to the chem film reaction. A part that's only cleaned (degreased) but not **deoxidized** still carries that oxide plus **alloying "smut"** (copper, iron, silicon, magnesium residues the cleaner leaves behind, especially on 2024/7075). On that surface the chem film forms thin, blotchy, or not at all. The **deox** dissolves the oxide and smut and leaves bright, reactive aluminum — exactly what the chem film needs. **This is the single biggest difference from the zinc chromate line**, where the equivalent step is a quick acid "activation"; on aluminum the deox is a heavier-duty, non-negotiable station.

★ **REMEMBER**

Clean removes the grease; deox removes the oxide and smut. You need *both*, in that order. Skipping or shortchanging the deox is the #1 cause of no-color/patchy chem film. No deox, no film.

☀ **FUN FACT**

Aluminum's oxide re-grows in *seconds* after deox — which is why the line keeps moving and why you don't let deoxidized parts sit around dry before the chem film. The fluoride in many chem film baths actually keeps nibbling that re-forming oxide away during the dip so the reaction can keep going. Aluminum is always trying to re-armor itself; the chemistry just stays a step ahead.

CHAPTER 5

WHY THE ORDER MATTERS

EACH STATION SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY — AND BY SAFETY

You might think the station order is just convention. It isn't. Each station exists *because of* the one before and after it.

RECEIVE FIRST — BECAUSE CONDITION AND ALLOY MATTER

Because chem film forms differently on different alloys, and a corroded, anodized, or wrong-metal part can't be saved by chem film. You confirm it's genuinely **bare aluminum**, the right alloy/temper for the print, and in good condition *before* any chemistry is wasted on it.

CLEAN BEFORE DEOX — BECAUSE ACID CAN'T CUT THROUGH GREASE

Oils and shop soil block the deoxidizer from reaching the oxide. The **alkaline clean** removes grease, fingerprints, and machining fluids so the deox can do its real job. Skip it and the deox works unevenly, leaving patches of oxide behind.

> ★ **REMEMBER**
>
> **Grease first, oxide second.** A degreaser can't strip oxide, and a deoxidizer can't cut through grease. Run them in order or you get a half-treated surface.

DEOX BEFORE CHEM FILM — BECAUSE OXIDE BLOCKS THE REACTION

This is the locked, critical step. The **deox** strips the tenacious aluminum oxide and alloying smut, leaving fresh, reactive metal. On that surface the chem film forms evenly and fully. On an oxidized surface it forms thin, blotchy, or not at all.

> ★ **REMEMBER**
>
> **No deox, no chem film.** A fully deoxidized aluminum surface is what lets the film form an even, full-coverage, full-protection coat. Deox right before you chem-film — an aluminum surface re-oxidizes in seconds to minutes.

RINSE BETWEEN EACH CHEMISTRY — BECAUSE DRAG-IN SHIFTS THE BATHS

Each chemistry is its own — alkaline cleaner, acidic deox, acidic chem film. Carry a film of one into the next and you **drift the next tank's pH and contaminate it.** The rinses are the firewalls.

> ★ **REMEMBER**
>
> *"Drag cleaner into the deox, or deox into the chem film, and you blunt both."* Rinses keep each chemistry out of the next tank. These are your firewalls on the line.

CHEM FILM — THE MAIN EVENT (COOL/MILD, A FEW MINUTES)

Now — and only now, with a clean, fully deoxidized aluminum surface — the chem film actually forms. The step is **cool or mildly warm, acidic, and a *few minutes*** (longer than the zinc chromate's seconds). This is the payoff stage and, on Type I, the **carcinogen stage**; treat it with total respect.

> ⚠ **HEADS-UP**
>
> On **Type I**, the chem film solution is **hexavalent chromium — a confirmed human carcinogen — and acidic.** Even at a few minutes of immersion, the mist, the drag-out, and any splash are all Cr(VI). Full PPE, ventilation, and mist suppression every time. (Full detail at Station 06.) On **Type II**, the hazard is lower but it's still an acidic process — goggles, gloves, apron.

RINSE AFTER CHEM FILM — GENTLY, AND COOL

The part leaves wet with chem film solution that must come off — but the **fresh film is a soft, soluble gel.** A rinse that's too hot, too long, or too aggressive will **leach the soluble chromium back out** (on Type I, the self-healing reservoir) and weaken both color and protection. The rinse must be **cool and controlled** — enough to wash off drag-out, not so much that it strips the film.

> ★ **REMEMBER**
>
> This rinse is a balancing act: **wash off the (Type I carcinogenic) drag-out, but don't wash away the protection.** Cool water, short dwell, no hot final rinse. Skimp and you ship drag-out; overdo it (hot/long) and you strip the film.

THEN DRY COOL, THEN INSPECT

- **Dry low and gentle** so the gel **cures** instead of cracking. (The single biggest operator mistake on this line lives here — Station 08 and Part Three.)
- **Inspect** to confirm color/coverage and — for electrical jobs — **contact resistance.**
- Chem film usually needs **no seal** — it's left bare or sent straight to **primer/paint**, which it's designed to bond.

> ⚠ **HEADS-UP — THE ORDER *AND* THE TEMPERATURE ARE LOCKED**
>
> You **dry cool** because the freshly formed film is a **hydrated gel.** Dry it in a hot oven (like an anodize seal or a paint bake) and you **dehydrate and craze the film, drive off the soluble Cr^{6+} (Type I), and destroy corrosion resistance and self-healing.** On this line, "more heat to dry faster" is a coating-killer. Patience is part of the procedure.

> ⚙ **TECHNICAL STUFF**
>
> The recurring villain that justifies the rinses is **drag-out** (solution clinging to a part as it leaves a tank) and its mirror, **drag-in** (that film contaminating the next tank). On a **Type I** line drag-out is doubly serious: it wastes chromium *and* spreads a carcinogen around the shop and into the wastewater. That's why chem film lines use **dragout-recovery rinses** and careful Cr(VI) waste treatment (Part Three).

CHAPTER 6

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND ACIDS — AND, ON TYPE I, A CARCINOGEN

A chem film line works with **alkaline cleaners, acidic deoxidizers, and — on Type I — a carcinogenic hexavalent-chromium bath and its mist.** PPE is your *last* layer of protection (after ventilation and mist suppression), but it is essential.

★ **REMEMBER**

There is no part, no order, and no deadline worth your health. Scrap can be reworked. Your lungs cannot. On a Type I line that's not a slogan — Cr(VI) makes it literal.

THE MASTER PPE SET (WEAR AT ALL CHEMICAL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; acids and chromic acid find them.
- **Face shield** — *over* the goggles for any work at the deox, chem film, or any brush/spray task.
- **Chemical-resistant gloves** — elbow-length, rated for chromic/nitric/fluoride-bearing chemistry per the SDS (nitrile, neoprene, or PVC). Inspect for pinholes before each use — a chrome ulcer starts at a single break. **Fluoride-bearing deoxidizers deserve special respect** (see Heads-Up).
- **Chemical-resistant apron/suit** — full coverage, chest to below the knee.
- **Chemical-resistant, non-slip boots** — floors around tanks and rinses are wet and slick.
- **Dedicated work clothing** — kept separate from street clothes; never worn home (keeps Cr(VI) out of your car and house on Type I lines).
- **Respiratory protection** — the respirator your shop's Cr(VI) program specifies for Type I tasks, and per the SDS for any misting acid task. Required whenever you could be exposed above the limits, and any time ventilation is degraded.

⚠ **HEADS-UP — HYDROFLUORIC / FLUORIDE HAZARD**

Many aluminum **deoxidizers contain fluorides** (and some use hydrofluoric acid or bifluorides). **Fluoride/HF burns are uniquely dangerous** — they can be deep, delayed, and serious even when the skin looks only mildly affected, and HF specifically requires **calcium gluconate** first aid per your SDS. Know whether your deox contains fluoride, where the calcium gluconate gel is, and the specific first-aid steps *before* you run it.

💡 **TIP**

Put gear on *in order*: boots and apron first, then gloves, then goggles, then face shield and respirator last. Take it off in reverse, and rinse your gloves *before* removing them. This keeps contaminated gloves away from your face — critical with a carcinogen (Type I) and with fluoride deox.

STATION	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Receive/Inspect	Mechanical; sharp/heavy parts	Cuts, pinches
02 Alkaline Clean	**Caustic/alkaline**	Skin/eye burns; slips
03 Rinse	Residual chemistry; wet floor	Mild irritant; **slips**
04 Deoxidize/Desmut	**Strong acid; often fluoride**	Burns; **fluoride/HF burns**; mist
05 Rinse	Residual acid; wet floor	Irritant; slips
06 Chem Film (Type I)	**Cr(VI) chromic acid + mist; acidic**	**Carcinogen; burns; respiratory harm**
06 Chem Film (Type II)	Acidic (trivalent/Cr-free)	Burns; irritation (lower hazard)
07 Rinse	(Type I) Cr(VI)-contaminated water	Carcinogen; burns; slips
08 Dry/Inspect	Warm air, hot parts; residual chemistry	Burns; mild irritant

⚠ **HEADS-UP**

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting Cr(VI) (Type I) and fluoride. Wash thoroughly before breaks and before leaving; keep food/drink out of the work area; keep work clothes separate from street clothes.

CHAPTER 7

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. **Locate, before your first shift:** the nearest **emergency eyewash** (within ~10 seconds of every chemical station), the **safety shower**, the **fire extinguisher** and alarm, the **spill kit**, the **SDS binder**, the **calcium gluconate gel** (if your deox contains fluoride/HF), and — on a Type I line — your shop's **Cr(VI) regulated-area** boundaries and decon station.

⚠ **HEADS-UP**

Eyewash and shower stations must always be unblocked. Never stack parts, totes, or boxes in front of them. The day you need one is the day you can't move boxes out of the way.

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Acid / chem film / deox on skin	Safety shower; flush **at least 15 minutes.** Remove contaminated clothing while flushing. **If the deox contains fluoride/HF, apply calcium gluconate per the SDS and get medical help fast.** Any chromic-acid contact on broken skin needs medical attention (chrome-ulcer risk). Bring the SDS.
In eyes	Eyewash; hold eyelids open and flush **at least 15 minutes**, rolling the eyes. Do not rub. Get medical attention immediately — these are corrosive to eyes. Bring the SDS.
Inhaled chromic-acid/acid mist, feel ill	Move to fresh air immediately. Report it — don't "tough it out." Seek medical attention for anything beyond brief mild irritation; report as a possible Cr(VI) exposure (Type I).
Cr(VI) / chemical swallowed	**Do not** induce vomiting unless the SDS directs it. Follow SDS first aid; call poison control / medical help immediately.
Nosebleed / sore nasal passages (Type I)	Report it — recurrent nosebleeds or nasal soreness can be an early sign of Cr(VI) over-exposure; it triggers a ventilation/exposure check. Don't ignore it.
Chemical spill	Alert others; keep people away. Clean only if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. **Never let chem film or acid spills reach a floor drain.**

⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

Mandatory for *all* electrical/mechanical maintenance — heaters, pumps, hoists, dryer, ventilation. Physically lock the power off and tag it so no one can re-energize while someone works. "I'll just be a second" is how people get hurt.

⚠ **HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**

When diluting acid or making bath additions per your procedure, **always add ACID/chemical to WATER — never water to concentrated acid.** Adding water to concentrated acid releases heat so violently it can erupt acid into your face. Memory aid: *"Do as you oughta — add acid to water."*

☀ **FUN FACT**

The 15-minute flush rule isn't arbitrary — it's the time emergency-response standards found necessary to dilute and carry away most corrosive chemistry before deep tissue damage. That's why eyewash and shower stations must deliver a steady, hands-free flow for a full 15 minutes — and why fluoride burns, which keep working after contact, need that *plus* the calcium gluconate antidote.

CHAPTER 8

THE SAFETY PHILOSOPHY OF A CHEM FILM LINE

A QUICK PROCESS IS STILL ACIDS — AND ON TYPE I, STILL A CARCINOGEN — TREAT IT THAT WAY

People sometimes assume that because chem film is fast, cool, and "just a dip," it's the safe one. That's a dangerous assumption. **The process is short, but the chemistry includes the most hazardous in the conversion-coating family.**

- **The Type I bath is a carcinogen.** Hexavalent chromium is a confirmed human carcinogen whose worst hazard — the mist — is invisible. Ventilation and mist suppression are primary controls, not afterthoughts.
- **The acids bite — and some are fluoride.** The deoxidizer and the chem film are strongly acidic; fluoride-bearing deox can cause uniquely dangerous burns.
- **The cleaner is caustic.** Alkaline burns and eye injuries are real.
- **Drag-out spreads the hazard.** Every part carries a film out of the tank. On Type I, sloppy handling spreads a carcinogen across the shop and into the rinses.
- **The waste is regulated.** Cr(VI) must be **reduced to Cr(III) and precipitated** before discharge; it can never go down a drain (Part Three).
- **The product is regulated.** RoHS/REACH mean Type I may be restricted for your market — know which Type your shop runs and why.

★ **REMEMBER**

The safety philosophy of this line is simple: **respect the acids, respect the carcinogen (Type I), control the mist, contain the drag-out, treat the waste, and keep your hands away from your face.** A chem film line isn't "the quick easy one" — it's an *acid (and, on Type I, Cr VI)* line that happens to be quick, and it deserves the same discipline you'd give any tank of corrosive, carcinogenic chemistry.

★ **REMEMBER**

You've now got the whole foundation: what a conversion coating is, what makes chem film special (two jobs — protect/paint-base *and* conductivity; plus Type I self-healing), how the eight-station line works as one connected system, why the order is locked by chemistry (especially the critical deox), the Cr(VI) hazard and the RoHS driver, what to wear, what to do in an emergency, and the safety mindset that ties it together. Turn the page knowing the *why* — and the *how* will make a lot more sense.

PART TWO — THE LINE, STATION BY STATION (STATIONS 01–08)

STATION 01 OF 08

RECEIVE & INSPECT

"YOU CAN'T CHEM-FILM ALUMINUM THAT'S THE WRONG ALLOY, ANODIZED, OR ALREADY CORRODED."

Before any chemistry touches a part, someone confirms it arrived as **bare aluminum**, the right alloy/temper, in the right condition, with the right Type/Class called out. This feels like the un-glamorous step. It isn't: chem film forms differently on different alloys, and it won't form at all on an already-anodized or non-aluminum surface.

• WHAT YOU'RE DOING & WHY IT MATTERS

- **Confirm the job.** Right part number, quantity, spec. Check the traveler for **Type (I/II) and Class (1A/3)**, the **alloy** (6061? 2024? 7075? cast?), masking requirements, and whether it's going to **paint** afterward or staying bare/electrical.
- **Confirm the substrate.** It must be **bare aluminum.** Anodized, painted, steel, or zinc-plated parts don't belong here — flag them.
- **Inspect condition.** Heavy corrosion/pitting, deep stains, or weld/heat discoloration are warning signs. Chem film is a thin film — it won't bury heavy damage.

• OPERATOR ESSENTIALS

ITEM	GUIDANCE
Substrate check	**Bare aluminum** = good. Anodized / painted / steel / zinc = **wrong line — flag it**
Alloy / temper	Confirm vs traveler — high-Cu alloys (2024/7075) need thorough desmut; castings may need fluoride deox
Type / Class	Confirm **Type I or II** and **Class 1A or 3** before you start
Going to paint?	Affects handling and whether bare color/coverage is the deliverable
Handling	Clean gloves — fingerprints and oils become defects under the film
Masking	Confirm areas to be left uncoated/threaded/ground per print

⚠ **HEADS-UP**

Mechanical hazards rule this station — sharp machined edges, heavy fixtures, pinch points. Wear cut-resistant gloves for handling, lift with your legs, and get help for heavy or awkward parts.

• STEP-BY-STEP PROCEDURE

1. **Read the traveler.** Confirm part, quantity, alloy, **Type/Class**, masking, and the after-process route (paint vs bare/electrical).
2. **Check substrate and condition.** Bare aluminum, right alloy. Set aside anything anodized, corroded, or wrong-metal for your lead.
3. **Handle with clean gloves.** Avoid bare-hand contact — skin oils cause spotty film.
4. **Verify masking.** Threads, bores, bonding pads, or surfaces that must stay uncoated should be masked per print.
5. **Flag anything unusual** (wrong substrate, heavy corrosion, mixed alloys/lots) before the load enters the cleaner.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Running the wrong substrate** (anodized/steel) without flagging it.
2. **Bare-handed handling** — fingerprints become film voids and paint-adhesion failures.
3. **Ignoring alloy** — treating a high-copper 2024 part like a 6061 part and under-desmutting it.
4. **Missing masking** — coating threads or bonding pads that had to stay bare.
5. **Processing visibly corroded parts** as if a thin film could hide the damage.

• TEACH-BACK CHECKLIST

- [ ] Explain why **substrate and alloy** matter before chem film.
- [ ] State which substrates belong here (bare aluminum) and what to do with anodized/steel/zinc (flag it).
- [ ] Identify conditions that need flagging (corrosion, anodize, wrong alloy).
- [ ] Describe why clean-glove handling and correct masking matter.

SIGN-OFF — STATION 01

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm I verified the job, substrate/alloy, Type/Class, masking, and that parts were bare aluminum in good condition, handled cleanly."

STATION 02 OF 08

ALKALINE CLEAN / DEGREASE

"GET THE GREASE OFF — SO THE DEOX CAN GET THE OXIDE OFF."

This is a dip (or spray) in an **alkaline cleaner** — usually a **mild, non-etch (or lightly inhibited) cleaner** — that removes oils, fingerprints, machining fluids, and shop soil from the aluminum. A clean surface is what lets the *deox* reach and strip the oxide. Some lines instead use a controlled **caustic etch** clean (which etches a little aluminum and brightens), always followed by a desmut/deox.

• WHAT YOU'RE DOING & WHY IT MATTERS

Grease and soil block both the deoxidizer and the chem film. The cleaner **wets the whole surface evenly** and lifts the contamination so everything downstream works uniformly. A poorly cleaned part shows up as **patchy film** two stations later.

⚙ **TECHNICAL STUFF**

Aluminum cleaners are usually **mildly alkaline and either non-etch (inhibited so they clean without attacking the metal) or mild-etch.** Strongly caustic etch cleaners *do* remove aluminum (and leave heavier smut that the deox must then strip), so they're chosen deliberately — for cosmetic brightening or to remove a damaged surface — not by accident. A good water-break-free surface (water sheets evenly, doesn't bead) is the classic sign the part is clean. **Concentration, temperature, and time come from the supplier TDS.**

• OPERATOR ESSENTIALS

PARAMETER	RANGE (VERIFY TDS)	NOTES
Cleaner type	`Mild alkaline, non-etch (or mild-etch)`	Non-etch protects dimensions/cosmetics
Temperature	`Often warm (~120–160°F typical)`	Heat helps cleaning; per TDS
Time	`A few minutes (per TDS)`	Long enough to wet/clean fully
Check	`Water-break-free after rinse`	Beading water = still contaminated

⚠ **HEADS-UP — ALKALINE/CAUSTIC**

The cleaner is **alkaline (caustic)** and can cause **skin and eye burns.** PPE every time: sealed goggles, face shield, chemical gloves, apron, boots. Caustic on skin feels "slippery/soapy" — that's it dissolving you; flush immediately and thoroughly.

STEP-BY-STEP PROCEDURE

1. **Check the cleaner** — concentration, temperature, and level in spec.
2. **Immerse/spray for the specified time**, with agitation as equipment allows.
3. **Withdraw and drain** back to the tank.
4. **Move promptly to rinse** — don't let cleaner dry on the part.
5. **Verify cleanliness** at the next rinse (water-break-free).

TOP MISTAKES NEW OPERATORS MAKE

1. **Running the cleaner cold or weak** — grease survives, film goes patchy.
2. **Skipping the water-break check** — shipping a “clean” part that wasn't.
3. **Letting cleaner dry** on the part (streaks/stains).
4. **Skipping PPE** because “it's just soap” — it's caustic.
5. **Using the wrong cleaner for the alloy** (over-etching a cosmetic part).

QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water beads after rinse	Incomplete cleaning; weak/cold cleaner	Titrate/heat per TDS; lengthen time
Etched/dull surface (unwanted)	Cleaner too aggressive for alloy	Switch to non-etch or reduce strength/time
Patchy film downstream	Oils/fingerprints left on	Improve cleaning; enforce glove handling

SIGN-OFF — STATION 02

Trainee: ____________________ Trainer: ____________________ Conc./Temp: __________

“I confirm the cleaner was in spec, parts came out water-break-free, and I wore required PPE.”

STATION 03 OF 08

RINSE (POST-CLEAN)

“A RINSE ISN’T A REST STOP. IT’S A FIREWALL.”

You just cleaned the part — but it’s wet with alkaline cleaner. This station’s job is to **wash that cleaner off** before the part hits the **acidic deox**, so the alkaline chemistry doesn’t neutralize and contaminate the acid tank.

• WHAT YOU’RE DOING & WHY IT MATTERS

The cleaner clinging to the part is **drag-out** (leaving the cleaner) / **drag-in** (arriving at the deox). The rinse knocks it down by dilution. Carry alkaline cleaner into an acid deox and you **neutralize the acid and build up salts** — the deox stops working right.

💡 **TIP — COUNTER-FLOW RINSING**

The professional approach is a **counter-flow (counter-current) rinse**: parts move forward, water flows backward. Fresh water enters the last (cleanest) stage and overflows into the dirtier one, so the cleanest water the part sees is the last thing to touch it before the deox. Saves enormous water and gives the deox a clean start.

```
FROM         +-------------+  PARTS ->  +-------------+         TO
CLEAN  ->    |  RINSE 1    | ---------> |  RINSE 2    | ---->  DEOX
             |  (dirty)    |            |  (clean)    |
             |  alkali in  |            |  near-neutral|
             +-------------+            +-------------+
                  ^   overflow <----------    ^
                  +------ <- FRESH WATER FLOWS BACKWARD <- -----+
                                                        FRESH IN
```

⚠ **HEADS-UP**

Rinse water carries dragged-in cleaner and is mildly alkaline. Wear **goggles, gloves, and non-slip footwear** — the floor around rinses is always wet, and **slips are the #1 injury here.**

★ **REMEMBER**

Rinse thoroughly between the alkaline cleaner and the acid deox. These two chemistries fight each other; the rinse keeps them apart so each tank stays in spec.

• TOP MISTAKES

1. A tired, contaminated rinse that drags alkali into the deox.
2. Treating the rinse as optional “downtime.”
3. Letting parts dry between rinse and deox (water stains, uneven deox).

SIGN-OFF — STATION 03

Trainee: ____________________ Trainer: ____________________ Date: ____________

“I confirm the rinse was clean/flowing, parts were rinsed and moved promptly to the deox, and I wore required PPE.”

STATION 04 OF 08 · THE MAKE-OR-BREAK PRETREATMENT

DEOXIDIZE / DESMUT

“STRIP THE OXIDE AND THE SMUT — OR THE CHEM FILM WON’T FORM. PERIOD.”

This is the step that makes aluminum chem film work, and the one most different from the zinc chromate line. The **deoxidizer** is an **acidic** solution that dissolves aluminum’s tenacious natural **oxide** and the **alloying smut** (copper/iron/silicon/magnesium residues) left by cleaning — leaving bright, chemically reactive aluminum for the chem film.

• WHAT YOU’RE DOING & WHY IT MATTERS

Aluminum re-grows an invisible oxide the instant it sees air, and that oxide *blocks* the chem film reaction. High-copper alloys (2024, 7075) also leave dark **smut** after cleaning. The deox removes **both** so the chem film can form an even, full-coverage film. **Skip or shortchange the deox and you get no-color, patchy, or powdery chem film** — the #1 root cause of chem film defects.

⚙ TECHNICAL STUFF

Deoxidizers are commonly **acid blends** — historically **chromic-acid-based**, but most modern deox is **chrome-free**: typically **nitric acid, often with fluorides** (and iron(III)/ferric sulfate as an oxidizer) to handle copper-bearing smut and high-silicon casting alloys. The fluoride is what cuts the stubborn oxide and silicon. **Concentration, temperature, and time come from the TDS** — too weak/short leaves oxide behind (no film); too strong/long over-etches and roughens the surface.

• OPERATOR ESSENTIALS

PARAMETER	RANGE (VERIFY TDS)	NOTES
Chemistry	`Acidic deox (nitric ± fluoride; ferric)`	Often **chrome-free**; some legacy chromic deox exists
Temperature	`Ambient (per TDS)`	No heat usually needed
Time	`~1–5 min (alloy-dependent)`	High-Cu alloys need a thorough desmut
Result	`Bright, uniform, water-break-free Al`	Dark smut remaining = under-deoxed

⚠ HEADS-UP — STRONG ACID, OFTEN FLUORIDE

The deox is **acidic** and frequently **fluoride-bearing.** Fluoride/HF burns are **uniquely dangerous** — deep, delayed, and serious even when the skin looks only mildly affected. Know whether your deox contains fluoride; locate the **calcium gluconate**; wear sealed goggles, face shield, chemical gloves, apron, boots. Never mix the deox with other chemistries.

★ REMEMBER

The deox is not optional and not a quick “activation” — it’s a heavy-duty, make-or-break step. A part that isn’t fully deoxidized cannot take chem film, no matter how good the chem film tank is. If you see dark smut or dull patches after the deox, the part isn’t ready — re-deox or flag it.

• STEP-BY-STEP PROCEDURE

1. **Check the deox** — concentration, level, temperature in spec.
2. **Confirm the part is well cleaned** (water-break-free) — grease blocks the deox.
3. **Immerse for the specified time** with agitation as allowed; watch for a bright, uniform, smut-free surface.
4. **Withdraw and drain** back to the tank.
5. **Move promptly to rinse, then chem film** — deoxidized aluminum re-oxidizes within minutes if it sits.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Under-deoxing** (too weak/short, or skipping it) — no-color/patchy chem film.
2. **Letting deoxed parts sit** before chem film — re-oxidation ruins coverage.
3. **Under-desmutting high-copper alloys** — blotchy, dark film.
4. **Over-deoxing** — over-etched, rough, dimensionally affected surface.
5. **Underestimating fluoride** — skipping the calcium-gluconate-aware PPE/first-aid.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dark smut remains	Deox weak/short; high-Cu alloy	Titrate/refresh; lengthen; verify desmut for alloy
Dull/over-etched	Deox too strong/long	Reduce time/strength per TDS
No/patchy film downstream	Incomplete deox; parts sat	Re-deox; shorten transfer to chem film

★ **REMEMBER**

If a chem film job goes wrong, **read the part backward and suspect the deox first.** Most no-color and patchy-film problems are born here, not at the chem film tank — a clean, fully deoxidized surface is the foundation everything else stands on.

SIGN-OFF — STATION 04

Trainee: ______________________ Trainer: ______________________ Conc./Time: __________

"I confirm the deox was in spec, parts came out bright/smut-free, were moved promptly to chem film, and I wore required PPE (including fluoride-aware first aid)."

STATION 05 OF 08

RINSE (POST-DEOX)

“WASH OFF THE ACID — AND DON’T LET THE FRESH, BRIGHT ALUMINUM SIT AND RE-OXIDIZE.”

You just deoxidized — but the part is wet with acidic (often fluoride-bearing) deox. Rinse it off so it doesn’t drag into and contaminate the chem film tank — but **move quickly**, because freshly deoxidized aluminum re-grows its oxide in seconds-to-minutes.

• WHAT YOU’RE DOING & WHY IT MATTERS

- **Remove deox drag-out** so it doesn’t drift the chem film bath’s pH/chemistry (and so fluoride doesn’t build up where it shouldn’t).
- **Don’t dawdle** — a long pause here lets the oxide re-form and undoes the deox you just did.

💡 **TIP**
Counter-flow rinse again, and keep the dwell short. Some shops use a clean, flowing rinse and immediately transfer to chem film with minimal drain time, precisely because deoxed aluminum is so eager to re-oxidize.

⚠ **HEADS-UP**
Rinse water here carries **acidic, possibly fluoride-bearing** drag-out. Goggles, gloves, non-slip footwear; route to waste treatment, not the drain.

★ **REMEMBER**
Rinse clean, then move straight to the chem film. A deoxidized aluminum surface is at its most reactive right now — that’s exactly when the chem film wants it. Don’t let it sit.

• TOP MISTAKES

1. Letting deoxed parts sit/dry (re-oxidation → patchy film).
2. A tired rinse dragging acid/fluoride into the chem film.
3. Treating the rinse as downtime.

SIGN-OFF — STATION 05

Trainee: ______________________ Trainer: ______________________ Date: ______________

“I confirm the rinse was clean/flowing, parts moved promptly to chem film before re-oxidizing, rinse water routed to waste treatment, and I wore required PPE.”

STATION 06 OF 08 · THE MAIN EVENT

CHEM FILM — ALODINE / IRIDITE

COOL/MILD, A FEW MINUTES — THE PROTECTIVE, PAINTABLE, CONDUCTIVE FILM FORMS HERE

This is the heart of the line. With a clean, fully deoxidized aluminum surface, you apply the **chem film** — by **immersion, brush, or spray** — for a **few minutes**, and the chromate conversion film grows. On **Type I** it's also the **carcinogen** step; treat the part — and yourself — with total respect.

• WHAT YOU'RE DOING & WHY IT MATTERS

The chem film solution reacts with the bare aluminum: the acid attacks the metal, the local pH rises, fluoride keeps the oxide from re-forming, and a **gel-like chromate/aluminum-oxide film precipitates.** On **Type I**, it **traps soluble Cr^{6+}** that gives the film its gold color *and* its self-healing. The film is **thin enough to stay electrically conductive** — the property that sets chem film apart from anodize. Color and coating weight (and therefore the **Class**) track time, chemistry, and the target.

• OPERATOR ESSENTIALS

PARAMETER	TYPE I (VERIFY TDS)	TYPE II (VERIFY TDS)
Chemistry	**Cr(VI)** chromic acid + fluoride + ferricyanide	**Trivalent (TCP) / chrome-free**
pH	~1.5–2.0	~3.5–4.0
Temperature	Room temp	Room temp to mildly warm
Time	~1–5 min (Class-dep.)	Per TDS; brush 1–3 min
Color	Gold/tan (1A heavier, 3 lighter)	Clear/faint — verify coverage
Class target	1A heavier/longer; 3 thinner	Same concept; 3 keeps low resistance

⚠ **HEADS-UP — TYPE I: THE SINGLE MOST DANGEROUS TANK ON THIS LINE**

The Type I bath is **hexavalent chromium, a confirmed human carcinogen,** and **acidic.** The hazard is mostly the **invisible mist/aerosol. Engineering controls first:** mist suppressant on the bath surface and local exhaust ventilation must be working *before* you run it. **Then** PPE: sealed goggles, face shield, chromic-acid-rated gloves, apron, boots, and the respirator your Cr(VI) program specifies. If mist is visible or exhaust is weak, **stop and report it.** (Type II is lower-hazard but still acidic — goggles, gloves, apron.)

• STEP-BY-STEP PROCEDURE

1. **Verify controls first (Type I).** Confirm ventilation/exhaust is pulling and the **mist suppressant** is present. No controls = no run.
2. **Check the bath.** pH in spec, temperature correct, level/clarity normal.
3. **Confirm the part is freshly deoxidized and rinsed** — and not sitting/re-oxidized.
4. **Apply for the specified time** for the target Type/Class, with gentle agitation (immersion) or even, overlapping strokes (brush) / even passes (spray). Watch the color develop (Type I).
5. **Withdraw smoothly and let drag-out drain back into the tank** (immersion) — recovering chromium and keeping Cr(VI) contained.
6. **Move directly to the controlled rinse.** Don't let the soft film dry on the way (it'll spot) and don't let drag-out drip across the floor.

💡 **TIP — READ THE COLOR AS IT FORMS (TYPE I)**

On a gold Type I film you can watch the part go from bright aluminum to pale straw to full iridescent gold over the dwell. **Pull at the target color/Class.** Under-color = too short/too weak or under-deoxed (thin, under-protected); chalky/powdery = too long/too aggressive (over-attacked film). The color is your real-time gauge — *but only on Type I*; on Type II you must verify coverage another way.

⚠ **HEADS-UP — THE FRESH FILM IS SOFT**

What you just grew is a **hydrated gel**, not a hard coating. It's easily **wiped, scratched, or marked** until it cures. Handle gently from here to dry-off; don't stack, rub, or let parts bang together. It hardens over the next several hours to ~24 hours.

⚙ **TECHNICAL STUFF — WHY A FEW MINUTES, AND WHY NOT HOT**

The film-forming reaction needs enough dwell to build coating weight, but **over-immersion keeps attacking the aluminum faster than the film can stabilize**, giving a loose, powdery, low-adhesion film. Running cool/mild keeps the attack controlled and the film tight and **conductive**. This is the opposite of anodizing, where a heated tank and current build a thick *insulating* oxide — here you want a thin, tight, *conductive* chromate.

• DIALING THE CLASS (COLOR / COATING WEIGHT)

TARGET	ROUGHLY HOW YOU GET THERE	RESULT
Class 1A (max corrosion)	Full time/chemistry; Type I = full gold/tan	Heaviest film, best salt-spray, paint base
Class 3 (low resistance)	Shorter time / lighter film; Type I = light gold	Thin film, **low contact resistance** for bonds

★ **REMEMBER**

Heavier film = more corrosion protection but higher electrical resistance; thinner film = lower resistance but less corrosion margin. That tension is exactly why MIL-DTL-5541 splits **Class 1A (corrosion)** from **Class 3 (low resistance).** Match the film to the **Class on the print** — don't run a heavy gold where the job needs a thin, conductive Class 3 bond, and don't run a thin Class 3 where the job needs Class 1A corrosion life.

⚙ **TECHNICAL STUFF — CONDUCTIVITY VS CORROSION**

Contact resistance rises with coating weight. A Class 3 part has to pass a **maximum contact-resistance** test (commonly cited around **5,000 μΩ/in² as-applied** at a specified pressure, with a higher allowance after salt spray — **verify the exact figures in the current MIL-DTL-5541**). That's why Class 3 is deliberately thin. The art of chem film is getting *enough* film for corrosion while staying *under* the resistance ceiling — the deox quality and dwell time are your main levers.

• DRAG-OUT: THE CARCINOGEN LEAVING THE TANK (TYPE I)

Every part carries a film of solution out of the chem film tank. On Type I that **drag-out** is wasted chromium, a contamination source, and a carcinogen.

- **Let parts drain over the tank** before transferring.
- **Use the dragout-recovery rinse** if your line has one.
- **Never let drag-out drip across the floor or toward a drain.**

⚠ **HEADS-UP**

Cr(VI) drag-out that reaches a floor drain is a **reportable environmental release** and a worker-exposure hazard. The chromium must end up either back in the tank or in the **waste-treatment system** (reduce → precipitate, Part Three) — never in the sewer.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Running Type I with weak/absent ventilation or mist suppressant** — the cardinal Cr(VI) sin.
2. **Under-deoxed parts** reaching the chem film — no-color/patchy film (look upstream!).
3. **Over-immersing** — powdery, low-protection film.
4. **Letting the soft film dry or get rubbed** before the controlled rinse.
5. **Ignoring Class** — heavy film on a Class 3 electrical part (fails contact resistance) or thin film on a Class 1A part (fails salt spray).
6. **Trusting "looks gold"** on a Type II part that's nearly clear — verify coverage another way.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
No/weak color (Type I)	Under-deoxed; dip too short/weak; cold	**Check the deox first**; lengthen to target color; check pH
Powdery / rub-off film	Over-immersion; bath too aggressive; pH off	Shorten dwell; verify pH/temp per TDS
Patchy / uneven color	Poor clean/deox; nesting; uneven agitation	Improve pretreat; reload; check agitation
High contact resistance (Cl 3)	Film too heavy/thick	Reduce dwell; lighter film per TDS
Visible mist over tank (Type I)	Mist suppressant depleted; exhaust weak	**Stop. Restore mist suppressant/ventilation, then resume**

• TEACH-BACK CHECKLIST

- ☐ State that the **Type I** bath is **hexavalent chromium, a confirmed human carcinogen**, and name the **primary** controls (ventilation + mist suppression) *before* PPE.
- ☐ Explain chem film's **two jobs** (protect/paint-base *and* stay conductive) and the **Class 1A vs Class 3** difference.
- ☐ Explain how **color relates to coverage/Class** on Type I, and why **Type II coverage must be verified** differently.
- ☐ Describe how **self-healing** works on Type I and why over-rinse/over-dry harms it.
- ☐ Describe correct **drag-out** handling and why Cr(VI) can never reach a drain.

⚠ **HEADS-UP — SAFETY GATE**

No operator runs the **Type I** chem film tank solo until the **Cr(VI) program / PPE / ventilation** row of the Training Matrix is signed. This is the carcinogen station — safety qualification comes first, always.

SIGN-OFF — STATION 06

Trainee: ______________ Trainer: ______________ Type/Class: ______ pH/Temp: ______

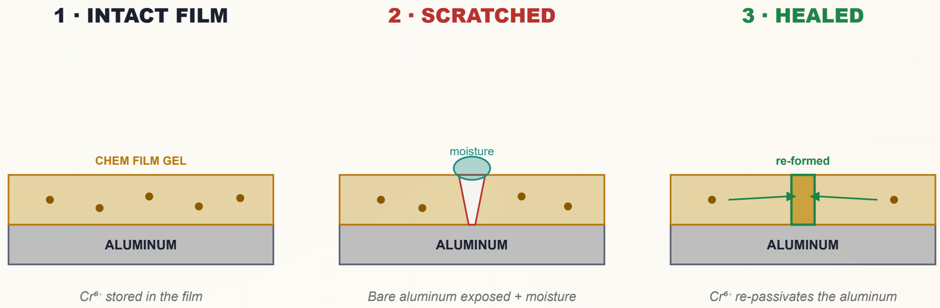
"I confirm ventilation and mist suppression were working (Type I), the bath was in pH/temperature spec, the part was fully deoxidized, I applied to the target color/Class for the correct time, handled drag-out correctly, and wore full PPE."

STATION 06 · THE SIGNATURE PROPERTY (TYPE I)

HOW THE FILM PATCHES A SCRATCH

ACTIVE CORROSION PROTECTION — THE COATING THAT FIGHTS BACK

SELF-HEALING — HOW TYPE I CHEM FILM PATCHES A SCRATCH



Active protection — the coating fights back. This is what Type II (Cr-free) chemistries struggle to match.

★ **REMEMBER**

This picture is the whole reason **Type I** gold chem film survived in aerospace: **the film carries its own repair kit** — vital where a machined edge or a field scratch exposes bare aluminum. It's also the whole reason it's hazardous and regulated — that mobile Cr^{6+} doing the healing is the carcinogen. The property and the problem are the same molecule.

FUN FACT

On aircraft, this self-healing is a real maintenance advantage: a technician can **brush "Alodine" onto a scratch or a freshly drilled hole** in the field and restore protection without a full strip — the very repair scenario we cover in Part Three. Engineers have spent decades trying to match that "active" behavior in chrome-free chemistries; modern Type II products keep getting closer, but the mobile-Cr(VI) trick is hard to fully reproduce.

STATION 07 OF 08

RINSE (GENTLE / COOL)

“WASH OFF THE DRAG-OUT — WITHOUT WASHING OFF THE PROTECTION.”

This rinse is a balancing act unique to conversion coatings. You must **remove the chem film drag-out** clinging to the part — but the fresh film is a **soft, soluble gel** (and on Type I it holds the self-healing reservoir). A rinse that’s too hot, too long, or too aggressive will **leach the soluble chromium back out**, weakening color and protection. So: **cool water, short dwell, gentle flow.**

• WHAT YOU’RE DOING & WHY IT MATTERS

- **Remove drag-out** so you don’t ship chemistry-wet parts or contaminate downstream (on Type I, so you don’t ship Cr(VI)).
- **Preserve the film** — keep the rinse cool and brief so the reservoir (Type I) stays in the coating.
- **Set up drying** — a clean, lightly rinsed surface dries spot-free (use DI water for the final rinse if your shop specifies it, to avoid water spots).

⚠ **HEADS-UP — NEVER A HOT FINAL RINSE ON A FRESH CHEM FILM**

A hot rinse does the same damage a hot oven does: it **leaches the soluble chromium and degrades the film.** Some operators use warm water “to help it dry.” Don’t. The dry-off (set *low*) does the drying — the rinse stays cool.

★ **REMEMBER**

Cool and brief. Enough to wash off the drag-out, not so much that you strip the film. On Type I, if you over-rinse you can literally watch the gold fade — that fade is your protection (and self-healing) leaving.

⚠ **HEADS-UP**

On Type I, this rinse water is **Cr(VI)-contaminated** — it goes to **waste treatment** (reduce → precipitate), never down a drain. Wear goggles, gloves, non-slip footwear.

• TOP MISTAKES

1. **Hot or prolonged rinse** — fades color, strips protection.
2. **Skimping** — leaves drag-out (Cr(VI) on Type I) on the part.
3. **Rough handling** of the still-soft film.

SIGN-OFF — STATION 07

Trainee: ____________________ Trainer: ____________________ Date: ____________

“I confirm I rinsed cool and briefly to remove drag-out without stripping the film, handled the soft coating gently, and routed Type I rinse water to waste treatment.”

STATION 08 OF 08 · MAKE-OR-BREAK + THE FINISH LINE

DRY-OFF (LOW TEMP) & INSPECT

"DRY IT GENTLY, OR YOU'LL UNDO EVERYTHING." — THE MOST IMPORTANT OPERATOR LESSON ON THIS LINE

This is where new operators ruin good parts. The freshly formed chem film is a **soft, hydrated gel.** It must be dried **gently and at low temperature** so it **cures** (slowly dehydrates and hardens) instead of **cracking.** Get this wrong and the corrosion resistance and (Type I) self-healing you grew at Station 06 are destroyed — invisibly. Then you do the final inspection.

PARAMETER	GUIDANCE
Temperature	**Low — below ~60°C (140°F)**; follow TDS (some specify lower)
Method	Warm-air dry, spin-dry, or ambient — gentle
Time	Long enough to dry, not so hot it bakes
What NOT to do	**No hot bake.** No anodize-seal or paint-bake oven on a fresh chem film

⚠ **HEADS-UP — THE RULE OF THIS LINE**

Do not over-dry / over-bake a fresh chem film. Over-drying **dehydrates and cracks the soft gel and destroys corrosion resistance** (and on Type I the self-healing). Keep dry-off **below your product's TDS limit (commonly ~60°C / 140°F).** MIL-DTL-5541 itself cautions against drying chem-filmed parts above that range. If your instinct from an anodize or paint line says "crank the oven," ignore it here.

• WHAT YOU'RE CHECKING (INSPECT)

CHECK	WHAT IT TELLS YOU	HOW
Color / coverage	Right finish; coverage present	Visual vs standard (Type I gold); deliberate check for clear Type II
Uniformity	No bare/thin spots	Visual, all faces
Rub / no-powder test	Film cured, adherent (not over-dipped/over-dried)	Light wipe — film should not rub off as powder
Contact resistance (Cl 3)	Electrical bond will work	Milliohm/contact-resistance meter per spec
No corrosion	Aluminum protected	Visual
Salt spray / adhesion	Corrosion life / paint grip	ASTM B117 / ASTM D3359 on samples (lab)

💡 **TIP**

The fast shop checks are **color/coverage** and a **light rub**. For **Type II**, color won't help — verify coverage per your shop's method. For **electrical (Class 3)** parts, the **contact-resistance reading is the real acceptance check**, not the look.

★ **REMEMBER**

Fresh film is soft; it hardens on aging. The coating keeps curing over the next several hours to ~24 h. Handle dried parts gently for the first day; don't pack hot, wet, or freshly coated parts tightly. If the part is going to paint, follow the spec's allowed window between chem film and paint.

SIGN-OFF — STATION 08

Trainee: ____________________ Trainer: ____________________ Dryer temp: __________

"I confirm I dried below the TDS limit, did not over-bake, verified color/coverage (and contact resistance for Class 3), and handled the curing parts correctly."

PART THREE — ACROSS THE LINE

CHOOSING THE RIGHT CHEM FILM

TYPE & CLASS SELECTION

TYPE = THE CHEMISTRY; CLASS = THE JOB. READ BOTH OFF THE PRINT.

Every chem film job is defined two ways at once. Get either one wrong and the part fails.

	TYPE I (HEX)	TYPE II (TRIVALENT/CR-FREE)
Chemistry	Cr(VI)	Cr(III) / chrome-free
Color	Gold/tan	Clear/faint
Self-healing	Strong	Weak/none
Corrosion	Highest (classic)	Good; modern close to Type I
Compliance	Restricted (RoHS/REACH)	Compliant

	CLASS 1A	CLASS 3
For	**Max corrosion** (painted/unpainted)	**Low electrical resistance**
Film	Heavier	Thinner
Acceptance	Salt spray / coverage	**Contact resistance** + corrosion

★ **REMEMBER**

A complete callout looks like **“Chem film per MIL-DTL-5541, Type II, Class 3.”** That’s *both* the chemistry (trivalent/Cr-free) *and* the job (thin, low-resistance for an electrical bond). Always confirm both halves before you run.

THE SELECTION LOGIC (PLAIN VERSION)

- **Needs an electrical ground/bond / RF-EMI continuity?** → **Class 3** (and confirm Type per market). Anodize is *not* an option here — it insulates.
- **Needs maximum corrosion or a paint/primer base?** → **Class 1A.**
- **For RoHS/REACH-governed commercial/electronics markets?** → **Type II.** Hex is not acceptable.
- **Aerospace/defense with a Type I qualification + self-healing requirement?** → **Type I** may be specified, with full Cr(VI) controls — but Type II (TCP) replacements are increasingly qualified; confirm the current print.
- **When in doubt** → treat Type I as restricted and **confirm the customer spec and current regulations.**

⚠ **HEADS-UP**

Don’t substitute Type or Class on your own authority. “It’s all Alodine” is wrong — a Type I Class 1A gold film on a Class 3 electrical bonding pad can **fail the contact-resistance requirement**, and a Class 3 thin film on a Class 1A part can **fail salt spray.** The print’s Type/Class is a real engineering requirement.

READING THE FILM

COLOR & COVERAGE VERIFICATION

ON TYPE I, COLOR IS YOUR GAUGE. ON TYPE II, YOU HAVE TO LOOK HARDER.

Because the film is so thin, you “read” Type I chem film mostly by **color**, backed by lab checks. **Type II is nearly colorless — so color won’t save you, and coverage verification becomes essential.**

TYPE / CLASS	LOOK	HOW YOU VERIFY COVERAGE
Type I, Class 1A	Full gold/tan iridescent	Visual color + uniformity; rub test; salt spray
Type I, Class 3	Light gold/straw (expected)	Visual + **contact resistance**
Type II (any)	Clear to faint blue	**Deliberate coverage check** — water-break/wetting, witness panels, or per-spec method; **don’t trust the eye**

★ **REMEMBER**

You cannot judge a Type II part “covered” just because it came out of the tank. The film is nearly invisible. Use your shop’s coverage-verification method (water-break-free wetting behavior, process control by time/chemistry on the deox + chem film, and periodic salt-spray/coverage witness panels). A clear film that *was never there* looks identical to a clear film that’s perfect.

⚙ **TECHNICAL STUFF**

Levers that move Type I color/weight (and Type II coating weight): **dip/brush time, bath concentration, pH, temperature, and — above all — how well the part was deoxidized.** More time/heavier chemistry → heavier, darker (Type I) film, but there’s a ceiling: **over-immersion stops building protective film and starts powdering it.** And remember the Class tension — heavier film raises contact resistance.

☀ **FUN FACT**

Type II’s near-invisibility is both its curse and its selling point: it’s *harder to inspect* but *cosmetically cleaner* — no gold tint on a finished aluminum surface that’s staying bare. Some manufacturers add a faint blue tint or a fluorescing dye to Type II products specifically so operators can *see* coverage under normal or UV light. Ask whether yours is tinted.

WHY CHEM FILM EXISTS

CONDUCTIVITY & CONTACT RESISTANCE

THE ONE THING CHEM FILM DOES THAT ANODIZE CANNOT — PROTECT ALUMINUM *AND* KEEP IT ELECTRICALLY LIVE

This is chem film's reason for being in electronics and aerospace electrical work. A finished aluminum surface usually needs *some* protection — but if it also has to carry a **ground, an RF/EMI bond, or a lightning path**, it must stay **electrically conductive.** Chem film does both. Anodize protects but **insulates.**

★ **REMEMBER**

Chem film is the conductive aluminum finish. Class 3 chem film is specified precisely where a protected aluminum surface must still make a low-resistance electrical contact — chassis grounds, connector flanges, waveguide/RF enclosures, faying surfaces, and bonding straps.

• HOW CONTACT RESISTANCE IS TESTED

A **contact-resistance (milliohm) meter** presses two contacts against the coated surface at a specified force and reads the resistance. MIL-DTL-5541 sets a **maximum** for **Class 3** — commonly cited around **5,000 microohms per square inch as-applied** (at a defined pressure such as 200 psi), with a higher allowance (often cited around 10,000 μΩ/in²) **after** salt-spray exposure. **Verify the exact numbers, pressure, and method against the current MIL-DTL-5541** — I'm giving the classic figures for orientation, not as a guaranteed spec value.

⚙ **TECHNICAL STUFF**

Resistance rises with coating weight, so there's an unavoidable trade-off: **enough film for corrosion vs. thin enough for low resistance.** That's the whole reason Class 3 exists as a separate class. Your levers to pass Class 3: **excellent deox** (a poorly deoxidized surface gives erratic, high readings), the **right Type/product**, and **controlled, shorter dwell** to keep the film light.

⚠ **HEADS-UP**

A Class 3 part that *looks* fine can still **fail contact resistance** if the film is too heavy or the surface was poorly deoxidized. For electrical parts, **the meter — not the eye — is the acceptance check.** Don't ship a bonding surface on looks alone.

☀ **FUN FACT**

On aircraft, this is literally a lightning-protection issue: the airframe must be electrically continuous so a strike can dissipate. Chem-filmed (Class 3) faying surfaces and bonding jumpers keep the structure bonded. Anodize there would create insulating gaps — exactly what you don't want when 30,000 amps are looking for a path.

APPLICATION METHODS

BRUSH, IMMERSE, OR SPRAY

SAME CHEMISTRY, THREE WAYS TO PUT IT ON — INCLUDING IN THE FIELD

Chem film is unusually flexible: you can apply it by **immersion** (tank), **spray**, or **brush/wipe** — and that last one makes it **repairable in the field**, a capability anodize doesn't have.

METHOD	WHERE USED	NOTES
Immersion	Shop line, full parts	Most uniform; best for Class 1A and full coverage; the process in Part Two
Spray	Large parts, structures	Even passes; capture overspray/runoff; ventilation critical (Type I)
Brush / wipe	**Field touch-up, repair, local areas**	"Alodine pen"/brush kit; restores scratched/drilled/machined spots

• FIELD TOUCH-UP (AIRCRAFT REPAIR)

A technician can **brush chem film onto a scratch, a reworked area, or a freshly drilled hole** to restore corrosion protection and the electrical bond — without stripping and re-coating the whole part. This is a major reason chem film is the maintenance-friendly aluminum finish.

⚠ **HEADS-UP — BRUSH/FIELD CR(VI) HAS ITS OWN EXPOSURE & CONTAINMENT RULES**

Brush-applying a **Type I (Cr VI)** product **outside a ventilated tank line** — on a hangar floor, inside a structure, in a confined space — can put a carcinogen right in the breathing zone with **little or no engineering ventilation.** That scenario demands: the **respiratory protection and PPE your Cr(VI) program specifies for brush application**, **containment** (catch and collect all drips, wipes, and rinse — nothing to the floor, ground, or drain), **collection of contaminated rags/applicators as hazardous waste**, and tight control in confined spaces. Where allowed, a **Type II (Cr-free) brush product** sharply lowers this hazard — many maintenance operations have switched to Type II touch-up for exactly this reason.

★ **REMEMBER**

The flexibility is real — **brush/spray/immerse all grow the same film** — but the **hazard controls travel with the chemistry, not the method.** Brushing Type I in the field is *still Cr(VI)*; if anything it's harder to control than a ventilated tank. Treat it with more care, not less.

WHY TYPE I GOLD EARNED ITS REPUTATION

THE SELF-HEALING MECHANISM

ACTIVE CORROSION PROTECTION — THE PROPERTY, AND THE PROBLEM, ARE THE SAME MOLECULE

The mechanism, step by step (Type I):

1. The cured film is a chromium/aluminum oxide-hydroxide matrix that **holds a reservoir of soluble hexavalent chromium.**
2. When the film is **scratched, drilled, or machined** down to bare aluminum and the spot gets **wet**, the trapped Cr^{6+} **dissolves into the moisture** at the damage.
3. That mobile Cr^{6+} **migrates to the exposed aluminum** and **re-passivates it** — chemically forming fresh chromate over the bare metal.
4. The damage is **re-protected**, without any operator action.

This is **active** corrosion protection (the coating responds to damage), versus **passive** barrier protection (anodize/paint just sits there).

★ **REMEMBER**

Self-healing = mobile Cr^{6+} leaching to re-protect damage. It's why Type I gives such strong protection at exposed edges and scratches so cheaply — and why **Type II (trivalent/Cr-free), which lacks that mobile Cr^{6+}, generally cannot fully match it.** This is the central trade-off in choosing Type I vs Type II for a demanding corrosion application.

⚙ **TECHNICAL STUFF**

The same mobility that heals scratches is the same mobility that makes the chromium **bioavailable and hazardous** — and the same reason **over-rinsing or over-drying leaches it out** and kills the property. Everything about Type I traces back to one fact: *the healing agent is soluble Cr(VI)*. That's the property, the hazard, and the regulatory problem, all in one molecule.

⚠ **HEADS-UP**

Because self-healing depends on the Cr^{6+} staying in the film, the upstream lessons connect: **don't over-rinse (Station 07), don't over-dry (Station 08).** Strip the reservoir and you keep the carcinogen risk while losing the benefit — the worst of both worlds.

THE TWO BIG DECISIONS ON ALUMINUM

CHEM FILM VS ANODIZE

PICK THE FINISH FOR THE JOB, THEN PICK THE COMPLIANT CHEMISTRY

• CHEM FILM VS ANODIZING

FACTOR	CHEM FILM	ANODIZE
How it's made	Chemistry only, **no current**	**Electric current** in acid
Thickness	Very thin	Thicker (esp. hardcoat)
Hardness/wear	Low	**High**
Corrosion	Good	**Higher** (esp. sealed/hardcoat)
Electrical	**Conductive (low resistance)**	**Insulating (dielectric)**
Paint/bond base	**Excellent**	Good (esp. unsealed)
Field repair	**Yes — brush touch-up**	No (needs re-anodize)
Speed/cost	Fast, low cost	Slower, higher cost

★ **REMEMBER**

Chem film when you need **conductivity, a paint/bond base, field-repairability, speed, and low cost** (and can accept lower corrosion/wear than anodize). **Anodize** when you need **maximum hardness, wear, and corrosion** and can accept an **insulating** surface. The conductivity question usually decides it.

• TYPE I VS TYPE II (THE ROHS DECISION)

- **Electronics (RoHS), EU/REACH-governed markets** → **Type II** (chrome-free/trivalent). Hex is restricted.
- **Aerospace/defense with a Type I qualification + self-healing requirement** → **Type I** may still be specified, with full Cr(VI) controls — but Type II (TCP) replacements are increasingly qualified.
- **When in doubt** → treat Type I as restricted and **confirm customer spec + current regs.**

★ **REMEMBER**

The honest summary: **Type I gold gives the best classic corrosion + self-healing and is easy to verify — but it's a Cr(VI) carcinogen and RoHS/REACH-restricted.** Type II is the compliant, lower-hazard, harder-to-see modern choice that's rapidly closing the corrosion gap. Both are legitimate to *know*; the direction of the industry is Type II.

⚠ **HEADS-UP**

Don't let "Type I is being phased out" make you complacent about a Type I tank you run *today.* If your shop runs gold (for a market/qualification that still allows it), it's still a carcinogen and still demands every Cr(VI) control in this manual.

KEEPING CR(VI) OUT OF THE ENVIRONMENT

CR(VI) WASTE TREATMENT

REDUCE, THEN PRECIPITATE — CHROMIUM NEVER GOES DOWN A DRAIN

Every drop of Cr(VI) from a **Type I** line — drag-out, rinse water, dumped baths, spills, contaminated rags from brush touch-up — must be treated before discharge. The classic two-step:

1. **Reduction ($Cr^{6+} \rightarrow Cr^{3+}$).** The wastewater is **acidified** (low pH) and a **reducing agent** (e.g., sodium metabisulfite/sulfite, or ferrous sulfate) converts the dangerous, soluble **hexavalent** chromium into far-less-hazardous **trivalent** chromium. You can often *see* it: the yellow Cr^{6+} shifts toward green/blue Cr^{3+}.
2. **Precipitation / removal (Cr^{3+} out of solution).** The pH is then **raised** (neutralized) so the trivalent chromium **precipitates as chromium hydroxide**, settled/filtered out as sludge for proper hazardous-waste disposal. Treated water is tested and discharged only within permit limits.

★ **REMEMBER**

Reduce then precipitate: $Cr^{6+} \rightarrow Cr^{3+} \rightarrow$ chromium-hydroxide sludge. The hazardous hexavalent form is destroyed first; only then is the chromium pulled out of the water. **Cr(VI) never, ever goes down a drain untreated** — illegal and an exposure hazard.

⚙ **TECHNICAL STUFF**

The reduction step needs **low pH**; the precipitation step needs **near-neutral/slightly alkaline pH.** So a chrome waste-treatment system is essentially a controlled pH-and-redox process. The end sludge is a regulated hazardous waste with its own manifest/disposal chain. **Note:** fluoride-bearing deox rinses also need their own treatment (fluoride precipitation, e.g., with calcium) — coordinate both waste streams with your facility's program.

• THE EASIER SIDE: TYPE II WASTE

A big practical advantage of **Type II**: there's **no hexavalent-chromium reduction step** to run, no Cr(VI) air scrubbing burden, and a smaller hazardous-waste footprint (still treat acids/fluoride/trivalent per permit, but no Cr(VI)). This easier, cheaper, lower-liability waste profile — on top of RoHS/REACH — is a major reason shops convert from Type I to Type II.

⚠ **HEADS-UP — AIR SIDE TOO**

The Type I tank exhaust carries Cr(VI) mist. A **scrubber** washes it out before discharge, under **air permits.** If the scrubber or exhaust is down, the tank isn't compliant *or* safe — stop and report it.

THE ONE RULE OPERATORS BREAK MOST

THE DON'T-OVER-DRY RULE

DRYING, AGING, AND THE SOFT GEL FILM — WHY HEAT IS THE ENEMY HERE

This deserves its own page because it's the most common way good chem film parts get ruined. **The fresh chem film is a hydrated gel.** Two things must happen, gently:

1. **Drying** — remove surface water so parts can be handled/packed/painted.
2. **Aging/curing** — over the next several hours to ~24 h, the gel slowly dehydrates and **hardens** into a coherent, durable, conductive film at room temperature.

Heat short-circuits this badly. Drying too hot (or rinsing too hot) **flash-dehydrates the gel**, causing it to **shrink, craze (microcrack), and lose its bound water and (Type I) mobile Cr^{6+}** — destroying corrosion resistance and self-healing. The killer: the damage is usually **invisible** — the part looks fine and fails salt spray (or shows erratic contact resistance).

DO

- Dry **below the TDS limit** (commonly ≤ ~60°C / 140°F).
- Use **warm air, spin-dry, or ambient.**
- Let parts **age/cure** ~24 h before rough handling.
- Trust **color + rub test + salt spray + contact resistance.**

DON'T

- **Bake** in a hot oven like an anodize seal or paint cure.
- Use a **hot final rinse** to "speed drying."
- Heavily **stack/abrade** fresh, soft parts.
- Assume "looks dry, looks fine" = good.

★ REMEMBER

Cool/mild process, cool rinse, gentle low-temp dry, then let it age. If you remember one *process* rule from this whole manual, make it this: **never over-dry a fresh chem film.** Heat is the enemy of a gel coating.

☀ FUN FACT

This is exactly backwards from the *anodize* line down the hall, where a hot seal and a bake are normal and *good*. Same metal — aluminum — opposite drying rule. It's a perfect example of why you run *each* chemistry by *its own* TDS and never carry a habit from one line to another.

HOW WE PROVE IT WORKED

QUALITY CHECKS

CHEM FILM IS JUDGED BY CORROSION LIFE, CONDUCTIVITY, COVERAGE, AND ADHESION — NOT JUST BY LOOKING RIGHT

CHECK	WHAT IT TELLS YOU	HOW
Color / coverage	Right finish; coverage present	Visual vs standard (Type I); **deliberate coverage check for clear Type II**
Uniformity	No bare/thin spots	Visual, all faces
Rub / no-powder test	Film cured, adherent (not over-dipped/over-dried)	Light wipe — film must not rub off as powder
Contact resistance (Class 3)	Electrical bond works	Milliohm/contact-resistance meter, per MIL-DTL-5541
Salt spray	Corrosion life	**ASTM B117**; 5541 Class 1A: withstand **168 hr** neutral salt fog with limited corrosion (verify exact criteria)
Paint adhesion (if painted)	Paint grips the film	**ASTM D3359** crosshatch/tape on painted samples
Coating weight / presence	Film actually formed	Lab methods per spec

★ **REMEMBER**

The real verdicts are **salt spray (ASTM B117)** for corrosion and, for electrical parts, **contact resistance.** Color/coverage and the rub test are fast **in-process predictors**; salt spray and the meter are the final judges. If salt spray fails but the part looked fine, suspect **incomplete deox, over-rinse, or over-dry.** If contact resistance fails, suspect **too-heavy film or poor deox.**

⚙ **TECHNICAL STUFF — WHAT "SALT SPRAY" IS**

Samples sit in a sealed cabinet sprayed with salt fog (ASTM B117); you record the *hours* until corrosion appears. MIL-DTL-5541 commonly requires **Class 1A** coatings to survive **168 hours** with no more than a small number of isolated corrosion spots in a defined area — **confirm the exact pass/fail counts and panel size in the current spec.** It's how customers specify and verify corrosion requirements.

💡 **TIP**

Keep retained samples, a **color standard** (Type I), and a **contact-resistance log** (Class 3). Run periodic salt-spray on production samples so you catch a drifting bath, a tired deox, a creeping rinse temperature, or a too-hot dryer *before* a customer does.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

SYMPTOM	LIKELY CAUSE(S)	FIRST THINGS TO CHECK
No / weak color (Type I)	**Under-deoxed surface**; dip too short/weak; cold; stale (re-oxidized) part	**Check the deox first**; lengthen dip; check pH/temp; shorten transfer time
Patchy / blotchy film	Poor clean or deox; copper smut (high-Cu alloy); fingerprints; nesting	Improve clean/desmut for alloy; enforce gloves; reload for full contact
Powdery / chalky / rub-off film	**Over-immersion**; bath too aggressive; pH off; **over-dried**	Shorten dwell; verify pH/temp per TDS; lower dryer temp
High contact resistance (Cl 3)	Film too heavy/thick; poor deox; corrosion/oxide on surface	Lighter/shorter film; improve deox; re-check after proper cure
Poor salt-spray (looks fine, fails)	**Incomplete deox**; over-rinse/over-dry stripped the film; wrong Class; thin film	Improve deox; cool/shorten rinse; **lower dryer temp**; run Class 1A per spec
Dried-out / crazed / cracked film	**Hot rinse or hot dry** dehydrated the gel	**Drop dryer temp below TDS limit**; eliminate hot final rinse
Can't verify coverage (Type II)	Film is nearly clear	Use coverage method (wetting/witness panels); consider tinted/UV product
Color/film won't repeat	Bath drifting (pH/chemistry/temp); deox tired; alloy/condition varies	Titrate/adjust per TDS (lab); maintain deox; standardize incoming parts

💡 **TIP**

The fastest diagnostic instinct: **read the part backward through the line.** No color or patchy? **Suspect the deox** (and the clean) first — it's the #1 chem film failure. Powdery? Over-immersion or over-drying. Fails salt spray but looks good? Incomplete deox or a hot rinse/dryer that stripped the film. High resistance? Film too heavy or poor deox. The defect usually names the stage that failed it.

⚠ **HEADS-UP**

Before you change bath chemistry, pH, temperature, or dryer settings, confirm with your lead and the supplier TDS. Adjustments and additions to a **Type I (Cr VI)** tank are made only by authorized personnel, in full PPE, with ventilation and mist suppression running.

YOUR DECODER RING

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

Alodine / Iridite: Common trade names for chem film (chromate conversion coating on aluminum).
Ambient: Room temperature; no heater.
Anodize: An *electrically driven* aluminum oxide finish — thicker, harder, and **insulating** (contrast with conductive chem film).
ASTM B117: The salt-spray (salt-fog) corrosion test; results in *hours* to corrosion.
ASTM D3359: Paint-adhesion (crosshatch/tape) test.
Chem film / chemical film: A thin chromate conversion coating grown on aluminum by chemistry alone — protects, primes for paint, and stays conductive.
Class (5541): The *job* — **1A = max corrosion**; **3 = low electrical resistance.**
Conversion coating: A film formed by chemically reacting (converting) the metal's own surface — no electric current.
Contact resistance: The electrical resistance across a coated surface; the **Class 3** acceptance check (microohms per square inch at a set pressure).
Cr(VI) / hexavalent chromium: Chromium in the +6 state; a confirmed human carcinogen; the active, self-healing, gold-coloring ingredient of **Type I.** OSHA 29 CFR 1910.1026.
Cr(III) / trivalent chromium: Chromium in the +3 state; far less hazardous; basis of **Type II** (TCP) chem film.
Deoxidize / desmut: The **critical acidic step** that strips aluminum's natural oxide and alloying smut so the chem film can form. No deox, no film.
Drag-out / drag-in: Solution clinging to a part leaving a tank (drag-out) and contaminating the next (drag-in) — on Type I, a carcinogen to contain.
Ferricyanide: Accelerator in Type I baths; speeds film growth and contributes to the gold color.
Fluoride / HF: Common deox ingredient; **fluoride/HF burns are uniquely dangerous** and need calcium gluconate first aid.
MIL-DTL-5541: Master spec for the *applied* chem film coating — Type I/II, Class 1A/3.
MIL-DTL-81706: Spec for the *chemical materials* used to make chem film.
AMS-C-5541: SAE/aerospace material-spec version of chem film.
PEL / Action Level: OSHA Cr(VI) air limits — **5 µg/m³ (PEL)** and **2.5 µg/m³ (action level)**, 8-hr averages.
pH: Acidity/alkalinity scale. Type I baths run ~1.5–2.0; Type II ~3.5–4.0 (verify TDS).
Reduce-then-precipitate: Cr(VI) waste treatment — reduce $Cr^{6+} \rightarrow Cr^{3+}$ (low pH), then precipitate Cr^{3+} as hydroxide (raise pH).
REACH / RoHS: EU/market rules restricting hexavalent chromium — the driver toward Type II.
Self-healing (active protection): The film's ability to re-protect a scratch as soluble Cr^{6+} leaches over bare aluminum — the signature of **Type I.**
Smut: Dark alloying residue (Cu/Fe/Si/Mg) left after cleaning, especially on 2024/7075; removed by desmut/deox.
Substrate: The base surface being coated — here, **bare aluminum / aluminum alloy.**
TCP: Trivalent Chromium Process/Pretreatment — a common Type II chemistry.
TDS / SDS: Technical Data Sheet (how to run a product) / Safety Data Sheet (its hazards). Follow both.
Type (5541): The *chemistry* — **Type I = hexavalent; Type II = trivalent/chrome-free.**

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

★ **REMEMBER**

An operator is "line-qualified" only when every station they run is signed. **Nobody runs the Type I chem film tank, brushes Type I in the field, or handles Cr(VI) waste solo until the Cr(VI) program / PPE / SDS row is signed.** Safety qualification comes first, always.

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT
1	**Cr(VI) program / PPE & SDS (Type I, all stations)**	____	______	______	____
2	**Fluoride/HF deox hazard & calcium-gluconate first aid**	____	______	______	____
3	Station 01 — Receive & Inspect (substrate, alloy, Type/Class)	____	______	______	____
4	Station 02 — Alkaline Clean / Degrease	____	______	______	____
5	Station 03 — Rinse (post-clean)	____	______	______	____
6	**Station 04 — Deoxidize / Desmut (the critical step)**	____	______	______	____
7	Station 05 — Rinse (post-deox)	____	______	______	____
8	**Station 06 — Chem Film (Type/Class, color/coverage, ventilation)**	____	______	______	____
9	Bath control — pH / chemistry checks	____	______	______	____
10	Station 07 — Rinse (cool/controlled, no leaching)	____	______	______	____
11	**Station 08 — Dry-Off (low-temp, don't over-dry) & Inspect**	____	______	______	____
12	Conductivity / contact-resistance testing (Class 3)	____	______	______	____
13	Brush / spray / field touch-up + containment	____	______	______	____
14	Cr(VI) waste treatment (reduce → precipitate)	____	______	______	____
15	Drag-out control / no Cr(VI) or fluoride to drain	____	______	______	____
16	LOTO (heaters/pumps/hoists/dryer/ventilation)	____	______	______	____
17	Emergency response — eyewash, shower, spill, calcium gluconate	____	______	______	____
18	Quality — salt spray / coverage / contact resistance	____	______	______	____

Operator name: ______________________ **Hire/start date:** ______________

Supervisor name: ______________________ **Review cycle:** ☐ Annual ☐ Other: __________

⚠ **HEADS-UP**

Safety rows (1, 2, 14, 15, 16, 17) are not optional and not "fill in later." An operator may not work *any* station — and especially not the Type I tank, field brush touch-up, or waste treatment — until the Cr(VI)/PPE/SDS, fluoride first-aid, waste, drag-out, LOTO, and emergency rows are signed. Safety competency comes before production competency, always.

PART FOUR — BACK MATTER & CERTIFICATION
20 QUESTIONS · PASSING SCORE 80% (16/20)

CHEM FILM COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**
Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q14, Q15) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ____________ **Score:** ________ / 20 **Safety-gate Qs (all correct): 5, 6, 14, 15**

Q1. A chem film conversion coating forms by:
A. An electric current depositing chromium from a bath (like anodizing) B. A chemical reaction between the cleaned/deoxidized aluminum surface and the process solution (no electric current) C. Spraying molten chromium onto the aluminum D. Baking a chromate powder onto the part in a hot oven

Q2. (Short answer) Name chem film's **TWO defining jobs** and explain why the second one makes it different from anodizing.

Q3. A **Type I (hexavalent)** chem film bath typically runs at what **pH**?
A. ~1.5–2.0 (strongly acidic) B. ~4.5–5.5 (mildly acidic) C. 7.0 (neutral) D. 11–13 (alkaline)

Q4. (Short answer) Where does the chem film step fit in the process, what **substrate** is it used on, and why is the **deox** step critical?

Q5. **[SAFETY GATE]** A **Type I** chem film bath is best described as:
A. A solution containing hexavalent chromium — a confirmed human carcinogen — regulated under OSHA 29 CFR 1910.1026 B. A harmless, food-safe rinse C. A non-hazardous trivalent-only solution D. An alkaline degreaser

Q6. **[SAFETY GATE]** OSHA's permissible exposure limit (PEL) for hexavalent chromium in air (8-hr average) is:
A. 500 µg/m³ B. 50 µg/m³ C. 5 µg/m³ D. 5 g/m³

Q7. (Short answer) Why must you **NOT dry a fresh chem film at high temperature**? What does over-drying do to the film?

Q8. Compared with **anodizing**, a chem film coating:
A. Is thicker and harder B. Is electrically insulating C. Cannot serve as a paint base D. Stays electrically conductive (low contact resistance), so the part can be grounded/bonded

Q9. (Short answer) What color is **Type I** vs **Type II** chem film, and why does **coverage verification matter more** on Type II?

Q10. Over-drying or over-baking a fresh chem film typically:
A. Makes it shinier and tougher B. Has no effect C. Cracks/dehydrates the soft gel film and degrades corrosion resistance (and Type I self-healing) D. Converts it into an anodize

Q11. (Short answer) Why must the **post-chem-film rinse** be *cool and brief*? What happens if it's hot or prolonged?

Q12. A **Type I** chem film's self-healing comes from:
A. Soluble hexavalent chromium (Cr^{6+}) in the film leaching out to re-passivate a scratch/cut edge B. A nickel underlayer C. Baked-on epoxy paint D. The aluminum substrate corroding protectively

Q13. Which regulations are the main driver to move from Type I (hex) to Type II (chrome-free/trivalent) chem film?
A. FDA food-contact rules B. FAA flight rules C. Building fire codes D. RoHS and REACH (they restrict hexavalent chromium)

Q14. (Short answer) [SAFETY GATE] List **four** pieces of PPE you must wear at the Type I chem film station, and name **one engineering control** that is your *primary* protection against the mist.

Q15. [SAFETY GATE] The correct waste treatment for hexavalent-chromium chem film rinse water/spills is to:
A. Pour it straight down the drain — it's dilute B. Reduce Cr^{6+} to Cr^{3+} (at low pH), then raise pH to precipitate the chromium as hydroxide for disposal C. Mix it with the alkaline cleaner and discharge D. Let it evaporate in an open tank

Q16. The RoHS/REACH-compliant replacement for Type I hexavalent chem film is:
A. Hard anodize B. Bare, uncoated aluminum C. A Type II trivalent / chrome-free chem film D. Iron phosphate

Q17. (Short answer) Give **two** honest tradeoffs between Type I and Type II chem film (one advantage of each is fine).

Q18. The spec that covers the **applied chromate conversion coating (chem film) on aluminum**, including Type I/II and Class 1A/3, is:
A. ASTM B117 B. ASTM B633 C. ASTM D3359 D. MIL-DTL-5541

Q19. A chem film conversion coating is applied to:
A. Bare carbon steel directly B. Freshly cleaned and deoxidized aluminum (and aluminum alloys) C. Cured powder coat D. Electroplated zinc

⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**
Questions 5, 6, 14, and 15 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q20. (Short answer) Name **two** quality checks used to verify a chem film finish (include at least one that applies to an electrical Class 3 part) and, in a few words, what each tells you.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

⚠ **HEADS-UP**

Keep this page out of the trainee's copy. Questions **5, 6, 14, and 15** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Self-check answer distribution (the 12 multiple-choice items): A = 3, B = 3, C = 3, D = 3. If your key clusters on one letter, you've mis-keyed.

Q1. B — A chemical reaction between the deoxidized aluminum surface and the solution; **no electric current.**

Q2. (1) Corrosion protection + an excellent paint/primer/adhesive bonding base; (2) it stays electrically conductive (low contact resistance). The second job is what separates it from **anodizing**, which is electrically **insulating** — chem film can protect aluminum *and* still carry a ground/bond.

Q3. A — ~1.5–2.0 (strongly acidic). (Type II runs milder, ~3.5–4.0.)

Q4. It's the **conversion/finishing step on bare aluminum** (after clean + deox). Substrate = **aluminum / aluminum alloys** (not steel or zinc). The **deox is critical because aluminum's tenacious natural oxide (and alloying smut) blocks the reaction** — no deox, no film.

Q5. A — A confirmed human carcinogen (hexavalent chromium); OSHA 29 CFR 1910.1026.

Q6. C — 5 μg/m³ (action level 2.5 μg/m³).

Q7. Because the fresh film is a **soft hydrated gel**; high heat **flash-dehydrates and cracks/crazes it, drives off the soluble Cr^{6+} (Type I), and destroys corrosion resistance and self-healing** — often invisibly (looks fine, fails salt spray). Keep dry-off below the TDS limit (commonly ≤ ~60°C / 140°F).

Q8. D — Stays electrically conductive (low contact resistance); anodize insulates.

Q9. Type I = gold/tan iridescent (color confirms coverage); **Type II = clear to faint** (nearly colorless). Because Type II is nearly invisible, **you can't judge coverage by eye** — coverage must be verified deliberately (wetting/witness panels/process control), or you can't tell a perfect film from no film.

Q10. C — Cracks/dehydrates the gel and destroys protection/self-healing.

Q11. Because the **fresh film is a soft, soluble gel** (on Type I it holds the self-healing Cr^{6+} reservoir). A **hot or prolonged rinse leaches the chromium back out**, fading color and weakening protection. Rinse cool and brief — enough to remove drag-out, not strip the film.

Q12. A — Soluble hexavalent chromium leaching out to re-passivate a scratch.

Q13. D — RoHS and REACH restrict hexavalent chromium.

Q14. PPE (any four): sealed chemical splash goggles, face shield, chromic-acid-rated chemical gloves, chemical apron/suit, chemical-resistant/non-slip boots, respirator per the Cr(VI) program. **Primary engineering control (any one):** local exhaust ventilation, fume/mist suppressant on the bath surface (and a scrubber). (Engineering controls are primary; PPE/respirator are last.)

Q15. B — Reduce $Cr^{6+} \rightarrow Cr^{3+}$ at low pH, then raise pH to precipitate chromium hydroxide for disposal. Cr(VI) never goes down a drain.

Q16. C — A Type II trivalent / chrome-free chem film.

Q17. Any two, e.g.: **Type I advantages** — best classic corrosion, strong self-healing, easy to verify by gold color. **Type I disadvantages** — Cr(VI) carcinogen, RoHS/REACH-restricted, heavier waste burden. **Type II advantages** — RoHS/REACH-compliant, far safer, easier waste. **Type II disadvantages** — weak/no self-healing, harder to verify (clear), historically slightly lower bare corrosion.

Q18. D — MIL-DTL-5541 (B633 is the zinc/chromate spec; B117 is salt spray; D3359 is paint adhesion).

Q19. B — Freshly cleaned and deoxidized aluminum / aluminum alloys.

Q20. Any two of: **color/coverage vs standard** (right finish/coverage — Type II needs a deliberate coverage check); **rub/no-powder test** (cured, adherent film); **contact resistance** (Class 3 — verifies the electrical bond will work); **salt spray ASTM B117** (corrosion life); **paint adhesion ASTM D3359** if painted. (At least one electrical-relevant check — contact resistance — should appear.)

★ **REMEMBER**

16/20 passes — but every safety question (5, 6, 14, 15) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people — especially on a Cr(VI) (Type I) line.

PRINT, FILL IN, SIGN

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CERTIFICATE OF COMPLETION

ALUMINUM CONVERSION COATING (CHEM FILM / ALODINE) TRAINING PROGRAM

This certifies that

____

OPERATOR NAME

has successfully completed the **Aluminum Conversion Coating (Chem Film / Alodine)** training course — covering all eight process stations (Receive & Inspect, Alkaline Clean/Degrease, Rinse, **Deoxidize/Desmut**, Rinse, **Chem Film/Alodine**, Controlled Rinse, and Low-Temperature Dry-Off/Inspect), chemical conversion coating without electric current, chem film's **two defining jobs** (corrosion + paint/bond base **and** electrical conductivity), the **conductivity differentiator vs anodizing**, **Type I vs Type II** and **Class 1A vs Class 3** selection, contact-resistance testing, brush/immerse/spray and field touch-up, the self-healing mechanism, the don't-over-dry rule, the **RoHS/REACH** driver, **Cr(VI) waste treatment** (reduce-then-precipitate), coverage verification, troubleshooting, and the safety practices for a **hexavalent-chromium (Cr VI) carcinogen** line — including ventilation, mist suppression, respiratory protection, fluoride/HF first aid, hygiene, drag-out control, and emergency response — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ____________________ **Re-certification due:** ____________________

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TRAINEE SIGNATURE

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CERTIFYING TRAINER

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SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer/aerospace specifications (MIL-DTL-5541, MIL-DTL-81706, AMS-C-5541), and applicable regulatory requirements before production use. Type I chem film contains hexavalent chromium, a confirmed human carcinogen restricted under RoHS and REACH — always consult current SDS and comply with OSHA 29 CFR 1910.1026, EPA, and local regulations.